EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction of Coordination Compounds

1. Which of the following statements is incorrect ?
(A) Coordination compounds and complexes are synonyms terms.
(B) Complexes must give ions in the solution.
(C) Complexes may give ions in the solution or may not give ions in the solution.
(D) Generally complex ion does not dissociate into its component parts even in the solution.

CCC093

2. Which will not give test of all the ions present in it
(A) $K_2Fe_2(SO_4)_4 \cdot 24H_2O$ (B) $(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$
(C) $K_3[Fe(CN)_6]$ (D) $Fe_2(SO_4)_3$

CCC094

3. Statement-I : $K_2SO_4 . Al_2(SO_4)_3 . 24H_2O$ is a double salt compound.
Because
Statement-II : It ionises to give a complex ion.
These questions contains, Statement-I (assertion) and Statement-II (reason).
(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
(B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
(C) Statement-I is True, Statement-II is False.
(D) Statement-I is False, Statement-II is True.

CCC100

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4. The number of ions formed on dissolving one molecule of $FeSO_4.(NH_4)_2SO_4.6H_2O$ in water is :
(A) 4 (B) 5 (C) 3 (D) 6

CCC107

Terminology Used in Coordination Compounds

5. The coordination number of Cr in the complex ion $[Cr(H_2O)_6]^{3\oplus}$ is :-
(A) Zero (B) 3 (C) 6 (D) None of these

CCC096

6. All ligands are :-
(A) Lewis acid (B) Lewis base (C) Neutral (D) None of these

CCC097

7. The co-ordination number of a central metal atom in a complex is determined by
(A) the number of ligands around a metal ion bonded by sigma and pi-bonds both
(B) the number of ligands around a metal ion bonded by pi-bonds
(C) the number of ligands around a metal ion bonded by sigma bonds
(D) the number of only anionic ligands bonded to the metal ion.

CCC105

8. Ligand in a complex salt are
(A) anions linked by coordinate bonds to a central metal atom or ion
(B) cations linked by coordinate bonds to a central metal or ion
(C) molecules linked by coordinate bonds to a central metal or ion
(D) ions or molecules linked by coordinate bonds to a central atom or ion

CCC106

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9. The total number of monodentate ligands attached to the central metal ion through coordinate (sigma) bond is called.

(A) Valency of the metal ion (B) Oxidation state of the metal ion

(C) Coordination number of metal ion (D) None of the above

CCC115

Formation Constant and Dissociation Constant

10. Which of the following complexes formed by $Cu^{2\oplus}$ ions is most stable ?

(A) $Cu^{2\oplus} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2\oplus}$, log K = 11.6

(B) $Cu^{2\oplus} + 4CN^{\ominus} \rightleftharpoons [Cu(CN)_4]^{2\ominus}$, log K = 27.3

(C) $Cu^{2\oplus} + 2en \rightleftharpoons [Cu(en)_2]^{2\oplus}$, log K = 15.4

(D) $Cu^{2\oplus} + 4H_2O \rightleftharpoons [Cu(H_2O)_4]^{2\oplus}$, log K = 8.9

CCC095

11. For the reaction of the type $M + 4L \rightleftharpoons ML_4$

(A) larger the stability constant, lower the proportion of ML_4 that exists in solution.

(B) larger the stability constant, higher the proportion of ML_4 that exists in solution.

(C) smaller the stability constant, higher the proportion of ML_4 that exists in solution.

(D) None of the above

CCC101

12. Consider the following reactions.

$$X + Y \rightleftharpoons XY \qquad K_1 = \frac{[XY]}{[X][Y]}$$

$$XY + Y \rightleftharpoons XY_2 \qquad K_2 = \frac{[XY_2]}{[XY][Y]}$$

$$XY_2 + Y \rightleftharpoons XY_3 \qquad K_3 = \frac{[XY_3]}{[XY_2][Y]}$$

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On the basis of reactions above which of the following is correct ?

(A) Overall stability constant = $K_1K_2K_3$

(B) $K_1K_2K_3 = \frac{[XY_3]}{[X][Y]^3}$

(C) $\text{Dissociation constant} = \frac{1}{\text{Formation constant}}$.

(D) All of the above are correct.

CCC103

Types of Ligands: On the basis of Charge

13. In the coordination compound $K_4[Ni(CN)_4]$, the oxidation state of nickel is –

(A) 0 (B) +1 (C) +2 (D) –1

CCC099

Types of Ligands: On the basis of Denticity; Ligand as Polydentate Ligand; Chelating Ligands; Ambidentate Ligands; Flexidentate Ligands;

14. The coordination number of cobalt in $[Co(en)_2Br_2]Cl_2$ is :-

(A) 2 (B) 4 (C) 6 (D) 8

CCC098

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15. Bidentate ligand is :-
(A) $CN^{\ominus}$ (B) Ethylene diamine (C) $EDTA^{4\ominus}$ (D) $SCN^{\ominus}$

CCC113

16. Triphenyl phosphine is :-
(A) Neutral and monodentate ligand (B) Neutral and tridentate ligand
(C) Uninegative and unidentate ligand (D) Trinegative and tridentate ligand

CCC114

17. Which of the following ligand does not act as bidentate ligand
(A) dipy (B) dien (C) $gly^{\ominus}$ (D) $dmg^{\ominus}$

CCC116

18. Diethylene triamine is:
(A) Chelating agent (B) Polydentate ligand
(C) Tridentate ligand (D) All of these

CCC117

19. $\overset{\ominus}{C}N$ is an ambidentate ligand because
(A) it has multiple bonds (B) carbon as a negative charge
(C) it form chelate (D) both C & N can donate lone pair

CCC118

20. Which of the following ligands may be ambidentate ligand –
(A) $NO_2^{\ominus}$ (B) $\overset{\ominus}{S}CN$ (C) Both (A) and (B) (D) None of these

CCC119

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21. An ambident ligand is one which
(A) is linked to the metal atom through two donor atom
(B) has two donor atoms, and only one of them has the capacity to form a coordinate bond [or a sigma (σ) bond]
(C) has two donor atoms, but either of the two can form a coordinate bond
(D) forms chelate rings.

CCC122

22. Which of the following does not form a chelate ?
(A) EDTA (B) Oxalate (C) Pyridine (D) Ethylenediamine

CCC124

23. Identify the correct statements for the behaviour of ethane-1, 2-diamine as a ligand.
(i) It is a neutral ligand. (ii) It is a bidentate ligand.
(iii) It is a chelating ligand. (iv) It is a unidentate ligand.
(A) (i), (ii) and (iii) (B) (ii) and (iii)
(C) (iii) and (iv) (D) (i), (iii) and (iv)

CCC125

24. The coordination number and the oxidation state of the element 'E' in the complex $[E(en)_2(C_2O_4)]NO_2$ (where (en) is ethylene diamine) are, respectively,
(A) 6 and 2 (B) 4 and 2 (C) 4 and 3 (D) 6 and 3

CCC104

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Organic Ligands

25. How many π electrons are donated by $C_5H_5^{\ominus}$ ligand -

(A) 2 (B) 4 (C) 5 (D) 6

CCC161

Werner Theory; Electrical and Molar Conductivity

26. Consider the following statements:

According the Werner's theory.

(a) Ligands are connected to the metal ions by covalent bonds.

(b) Secondary valencies have directional properties

(c) Secondary valencies are non-ionisable

(A) a, b and c are correct (B) b and c are correct

(C) a and c are correct (D) a and b are correct

CCC141

27. Which of the following complex will give white precipitate with barium chloride solution :-

(A) $[Cr(NH_3)_5Cl]SO_4$ (B) $[Cr(NH_3)SO_4]Cl$

(C) $[Co(NH_3)_6]Br_3$ (D) None of these

CCC142

28. A complex of platinum, ammonia and chloride produces four ions per molecule in the solution. The structure consistent with the observation is :

(A) $[Pt(NH_3)_4]Cl_4$ (B) $[Pt(NH_3)_2Cl_4]$

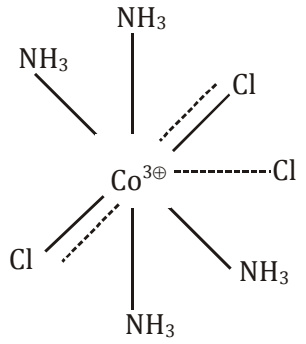
(C) $[Pt(NH_3)_5Cl]Cl_3$ (D) $[Pt(NH_3)_4Cl_2]Cl_2$

CCC143

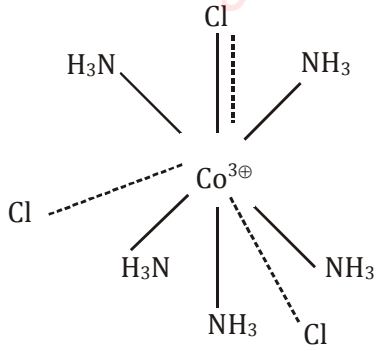
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29. Which of the following Werner's complex has least electrical conductivity ?

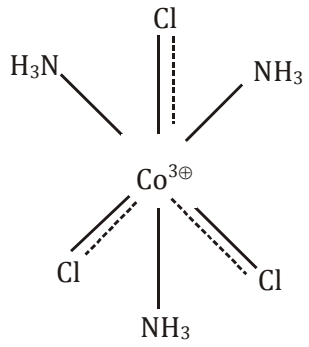
(A)



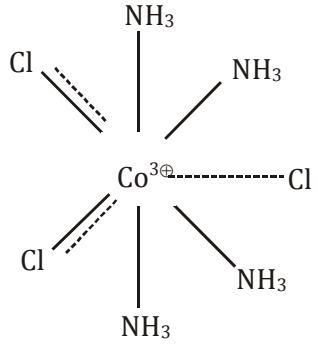
(B)



(C)



(D)



CCC144

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30. Which of the following has least conductivity in aqueous solution.
(A) $CoCl_3 . 4NH_3$ (B) $CoCl_3 . 3NH_3$ (C) $CoCl_3 . 5NH_3$ (D) $CoCl_3 . 6NH_3$
CCC145

31. Which gives only 50% mole of AgCl, when reacts with $AgNO_3$:-
(A) $PtCl_2 . 4NH_3$ (B) $PtCl_4 . 5NH_3$ (C) $PtCl_4 . 4NH_3$ (D) $PtCl_4 . 3NH_3$
CCC146

32. Number of electrolytes in the complex $CoCl_3 . 5NH_3$
(A) 3 (B) 2 (C) 4 (D) 5
CCC147

33. A co-ordination complex of cobalt has molecular formula containing five ammonia molecules, one nitro group and two chlorine atoms for one cobalt atom. One mole of this compound produces three mole ions in an aqueous solution. In reacting this solution with excess of silver nitrate solution, two moles of AgCl get precipitated. The ionic formula of this complex would be :-
(A) $[Co(NH_3)_4 . NO_2Cl] . [(NH_3)Cl]$ (B) $[Co(NH_3)_5Cl] . [Cl(NO_2)]$
(C) $[Co(NH_3)_5(NO_2)]Cl_2$ (D) $[Co(NH_3)_5] . [(NO_2)_2Cl_2]$
CCC148

34. What is the secondary valency of following compounds $PtCl_2.2NH_3$, $C_oCl_3.4NH_3$ and $NiCl_2.6H_2O$ if moles of AgCl precipitated per mole of the given compounds with excess $AgNO_3$ respectively are: 0, 1 and 2
(A) 6, 4, 4 (B) 4, 6, 6 (C) 4, 4, 6 (D) 2, 4, 6
CCC149

35. Correct formula of the compound which gives a white precipitate with $BaCl_2$ solution, but not with $AgNO_3$ solution is.
(A) $[Co(NH_3)_5Br]SO_4$ (B) $[Co(NH_3)_5SO_4]Br$ (C) $[Pt(NH_3)_4Cl_2]Br_2$ (D) $[Pt(NH_3)_4Br_2]Cl_2$
CCC150

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Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

36. The EAN of cobalt in the complex ion $[Co(en)_2Cl_2]^{\oplus}$ is :-
(A) 27 (B) 36 (C) 33 (D) 35
CCC154

37. If EAN of a central metal ion $X^{2\oplus}$ in a complex is 34. and atomic number of X is 28. The number of monodentate ligands present in complex are:-
(A) 3 (B) 4 (C) 6 (D) 2
CCC155

38. Which of the following does not obey Sidgwick effective atomic number rule -
(A) $Co_2(CO)_8$ (B) $[Fe(CO)_5]$ (C) $[V(CO)_6]^{\ominus}$ (D) $[Mn(CO)_6]$
CCC156

39. The complex that violates the EAN rule :-
(A) Potassium ferrocyanide (B) Potassium ferricyanide
(C) Tetra carbonyl Nickel (D) Cobalt (III) hexamine chloride
CCC157

40. The EAN of metal atoms in $Fe(CO)_2(NO)_2$ and $Co_2(CO)_8$ respectively are
(A) 34, 35 (B) 34, 36 (C) 36, 36 (D) 36, 35
CCC158

For More Join: @IITJEE_Advanced

41. Effective atomic number of $Co(CO)_4$ is 35, hence it is less stable. It attains stability by
(A) Oxidation of Co (B) Reduction of Co (C) Dimerization (D) Both (B) & (C)

CCC159

42. Among the following complexes which can act as oxidising agent.
(A) $[Mn(CO)_6]$ (B) $[Mn(CO)_6]^{\oplus}$ (C) $[Mn(CO)_5]^{\ominus}$ (D) $[V(CO)_6]$

CCC160

43. Ligands are neutral or ionic species capable of donating at least one electron pair to central metal. Hence ligands can be of different denticities.
$[Mn(CO)_5]$ can attain more stability by :
(A) Oxidation of itself
(B) Reduction of itself
(C) Dimerization
(D) Both (B) and (C)

CCC165

44. Ligands are neutral or ionic species capable of donating at least one electron pair to central metal. Hence ligands can be of different denticities.
Which of the following complexes follow Sidgwick EAN rule ?
(A) $[Fe(\eta^5 - C_5H_5)_2]$
(B) $K[PtCl_3(\eta^2 - C_2H_4)]$
(C) $[V(CO)_6]$
(D) $[Mn(CO)_6]$

CCC166

IUPAC Nomenclature of Coordination Compounds

45. $K_3[Fe(CN)_6]$ is :-
(a) Potassium hexacynoferrate(II)
(b) Potassium hexacynoferrate(III)
(c) Potassium ferricyanide
(d) Hexacynoferrate(III) potassium
Correct answer is :-
(A) Only (a) and (b) (B) Only (b) and (c) (C) Only (a) and (c) (D) Only (b) and (d)

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CCC172

46. The IUPAC name of $Fe(CO)_5$ is
(A) Pentacarbonyl ferrate(0)
(B) Pentacarbonyl ferrate(III)
(C) Pentacarbonyliron(0)
(D) Pentacarbonyliron(II)

CCC173

47. IUPAC name of $[Cr(NH_3)_4Cl_2]Cl$ is
(A) Tetraamminedichloridochromium(III) chloride
(B) dichloridotetraamminechromium(III) chloride
(C) trichloridotetraamminechromium(III)
(D) Chromiumtetraamminetrichloride

CCC174

48. The IUPAC name for the coordination compound $Ba[BrF_4]_2$ is
(A) Barium tetrafluorobromate (V)
(B) Barium tetrafluorobromate (III)
(C) Barium bis (tetrafluorobromate) (III)
(D) none of these

CCC175

49. Give the IUPAC name of the complex compound $[Co(NH_3)_4(H_2O)Br](NO_3)_2$.
(A) BromoaquatetraamineCobalt(III) nitrate
(B) BromoaquatetraaminoCobalt(III) nitrate
(C) BromoaquatetraammineCobaltate(III) nitrate
(D) TetraammineaquabromoCobalt(III) nitrate

CCC176

For More Join: @IITJEE_Advanced

50. The formula for the compound tris (ethane-1, 2-diamine)cobalt(III) sulphate is
(A) $[Co(en)_3]SO_4$ (B) $[Co(SO)_4(en)_3]$
(C) $[Co(en)_3](SO_4)_2$ (D) $[Co(en)_3]_2(SO_4)_3$

CCC178

51. The number of ions formed, when bis(ethane-1,2-diamine)copper(II) sulphate is dissolved in water will be
(A) 1 (B) 2 (C) 3 (D) 4

CCC179

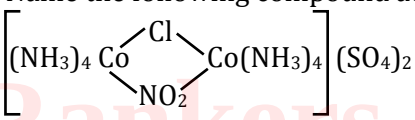
52. The IUPAC name for $[Co(NCS)(NH_3)_5]Cl_2$ is :-
(A) Pentaammine(thiocyanato-N)cobalt(III) chloride
(B) Pentaammine(thiocyanato-S)cobalt(III) chloride
(C) Pentaammine(isothiocyanato-N,S)cobalt(III) chloride
(D) Pentaammine(mercapto-N)cobalt(III) chloride

CCC181

53. The IUPAC name for $[Co(NH_3)_6]$ $[Cr(CN)_6]$ is :-
(A) Hexaamminecobalt(III) hexacyanochromate(III)
(B) Hexacyanochromium cobalt hexaammine(VI)
(C) Hexaammine cobalt(III) hexacyanochromium(VI)
(D) Hexacyanochromium(III) hexaammine cobalt(III)

CCC182

54. Name the following compound according to IUPAC system



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(A) Octammine-μ − chloro − μ-nitrodicobalt(II) sulphate
(B) Octaammine-μ − chloro − μ-nitrodicobalt(III) sulphate
(C) Octaammine-μ − nitro − μ-chlorodicobalt(III) sulphate
(D) None of the above

CCC186

IUPAC Nomenclature of Some Specific Complexes and Common Name

55. The IUPAC name of the Wilkinson's catalyst $[RhCl(PPh_3)_3]$ is
(A) Chloridotris(triphenylphosphine)rhodium(I)
(B) Chloridotris(triphenylphosphine)rhodium(IV)
(C) Chloridotris(triphenylphosphine)rhodium(0)
(D) Chloridotris(triphenylphosphine)rhodium(VI)

CCC180

56. The compound $Na_2[Fe(CN)_5NO^{\oplus}]$ is called
(A) Sodium pentocyanonitrosoniumferrate(II)
(B) Sodium nitroprusside
(C) Sodium nitrosoferrocyanide
(D) A & B both

CCC183

57. The correct IUPAC name of the complex $[Fe(\pi-C_5H_5)_2]$ is
(A) Cyclopentadienyliron(II) (B) Bis(η^5-cyclopentadienyl)iron(II)
(C) Dicyclopentadienylferrate(II) (D) Ferrocene

CCC184

For More Join: @IITJEE_Advanced

Structural Isomerism

58. Which of the following complex shows ionization isomerism.
(A) $[Cr(NH_3)_6]Cl_3$ (B) $[Cr(en)_2]Cl_2$ (C) $[Cr(en)_3]Cl_3$ (D) $[CoBr(NH_3)_5]SO_4$

CCC193

59. The compound $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_4Cl_2]Cl.2H_2O$ respresent
(A) Linkage isomerism
(B) Hydrate isomerism
(C) Ligand isomerism
(D) Ionisation isomerism

CCC195

60. Find the name of the hydrate isomer of $CrCl_3.6H_2O$, which is having lowest electrical conductivity excluding zero value of conductivity.
(A) Hexaaquachromium(III)chloride
(B) Tetraaquadichloridochromium(III)chloride dihydrate
(C) Pentaaquachloridochromium(III)chloride monohydrate
(D) Triaquatrichloridochromium(III)chloride trihydrate

CCC196

61. Which of the following compound cannot show coordination isomerism?
(A) $[Pt(NH_3)_4][PtCl_4]$
(B) $[Co(NH_3)_4Cl_2]\ [Cr(NH_3)_2(C_2O_4)_2]$
(C) $[Pt(NH_3)_4Cl_2]\ [Pt(SCN)_2Cl_2]$
(D) $[Pt(en)_2]^{2\oplus}$

CCC197

62. The type of isomerism present in pentaamminenitrochromium(III) chloride is :
(A) Coordination
(B) linkage
(C) hydrate
(D) Coordination position

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CCC198

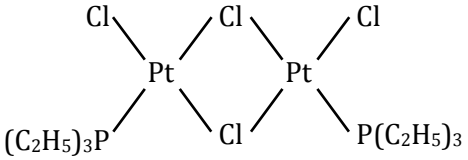
63. The two compounds sulphatopenta-amminecobalt(III) bromide and penta-amminebromo cobalt(III) sulphate represent :
(A) Linkage isomerism
(B) Ionisation isomerism
(C) Co-ordination isomerism
(D) Hydrate isomerism

CCC199

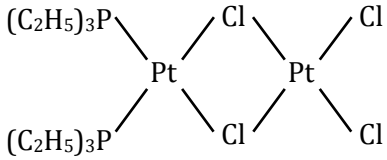
64. Which of the following complex may show atleast three type of isomerism, namely ionisation, hydrate and linkage :
(A) $[Co(H_2O)_5Cl]SO_4$
(B) $[Ag(NH_3)_2]\ [Ag(CN)_2]$
(C) $[Cr(H_2O)_5\ (NO_2)]SO_4$
(D) $[Ni(H_2O)_6]SO_4$

CCC200

65. The complexes given below show:



and



(A) Optical isomerism
(B) Co-ordination isomerism
(C) Geometrical isomerism
(D) Co-ordination position isomerism

CCC201

66. The type of isomerism present in pentaamminenitrochromium (III) chloride is :
(A) optical (B) linkage (C) hydrate (D) polymerisation

CCC221

For More Join: @IITJEE_Advanced

Stereoisomerism for Square Planar Complexes

67. The complex, [Pt(py)(NH_3)BrCl] will have how many geometrical isomers ?
(A) 3 (B) 4 (C) 0 (D) 2

CCC188

68. How many coordination isomers of [$Pt(NH_3)_4$][$PtCl_4$] show geometrical isomerism.
(A) All (B) One (C) Two (D) None

CCC202

69. How many stereoisomers are possible for the square-planar complex [$PtCl_3$ Br]$^{2\ominus}$
(A) 4 (B) 5 (C) 8 (D) None of these

CCC335

Stereoisomerism for Tetrahedral Complexes

70. The tetrahedral complex [M(A)(B)(X)(Y)], where A,B,X and Y are different ligands and M is a metal ion is
(A) optically inactive (B) rotate plane polarized light
(C) incomplete information (D) can't be said

CCC187

71. Tetrahedral complexes generally do not exhibit geometrical isomerism because :
(A) They have plane of symmetry.
(B) They have centre of symmetry.
(C) All positions about central atom are adjacent with respect to each other.
(D) All position about central atom are at 180° with respect to central atom.

CCC205

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72. Which of the following tetrahedral complex has two stereoisomers ?
[M = central atom, a, b, c, d = monodentate ligands]
(A) [Ma_2b_2] (B) [Ma_2bc] (C) [Mab_2c] (D) [Mabcd]

CCC206

73. A co-ordination compound [M abcd] has total two stereoisomers. The expected geometry of this complex is : [M = central atom, a, b, c, d = monodentate ligands]
(A) Square planar (B) octahedral (C) Tetrahedral (D) trigonal bipyramidal

CCC207

Stereoisomerism for Octahedral Complexes for Monodentate Ligand

74. Among the following coordination compounds/ ions
(i) [$Fe(CN)_6$]$^{3\ominus}$ (ii) [$Pt(NH_3)_2Cl_2$] (iii) [$Co(NH_3)_6$]$^{3\oplus}$ (iv) [$Cr(H_2O)_6$]Cl_3
Which species exhibit geometrical isomerism ?
(A) (ii) only (B) (i) and (ii) (C) (ii) and (iv) (D) (i) and (iii)

CCC210

75. Theoretically the Number of geometrical isomers expected for octahedral complex [Mabcdef] is:-
(A) Zero (B) 30 (C) 15 (D) 9

CCC215

76. Which of the following complex produce precipitates with $AgNO_3$ and exist in two geometrical isomeric form
(A) $PtCl_2.4NH_3$ (B) $PtCl_2.3NH_3$ (C) $PtCl_4.4NH_3$ (D) $PtCl_4.2NH_3$

CCC216

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77. The pair of molecules that exhibit geometrical isomerism are
(A) $[Cr(NH_3)_4Cl_2]^{\oplus}$and $[Co(NH_3)_3Cl_3]$
(B) $[Cr\ (NH_3)_5Cl]^{2\oplus}$ and $[Co(NH_3)_5Cl]^{2\oplus}$
(C) $[Cr(NH_3)_6]^{3\oplus}$ and $[Co(NH_3)_6]^{3\oplus}$
(D) All of the above show geometrical isomerism

CCC217

78. Which of the following complex shows optical isomerism -
(A) $[Cd(CN)_4]^{2\ominus}$ (B) $[Cr(H_2O)_3Cl_3].3H_2O$
(C) $[Zn(gly)_2]$ (D) $[Ni(dmg)_2]$

CCC218

79. Which of the following can exhibit geometrical isomerism ?
(A) $[MnBr_4]^{2\ominus}$ (B) $[Pt(NH_3)_3Cl]^{\oplus}$ (C) $[PtCl_2(P(C_2H_5)_3)_2]$ (D) $[Fe(H_2O)_5NOS]^{2\oplus}$

CCC219

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

80. Which of the following statements is/are true
(A) In $[PtCl_2(NH_3)_4]^{2\oplus}$ complex ion, the cis-form is optically active, while trans-form is optically inactive
(B) In $[Fe(C_2O_4)_3]^{3\ominus}$, geometrical isomerism does not exist, while optical isomerism exists
(C) $[Mabcd]^{\pm}$ square planar complexes exhibit both optical as well as geometrical isomerism
(D) In $[Mabcd]^{n\pm}$ tetrahedral complexes, optical isomerism cannot be observed

CCC203

81. The number of optical isomers possible for $[Cr(C_2O_4)_3]^{3\ominus}$ is :
(A) Three (B) Two (C) One (D) Four

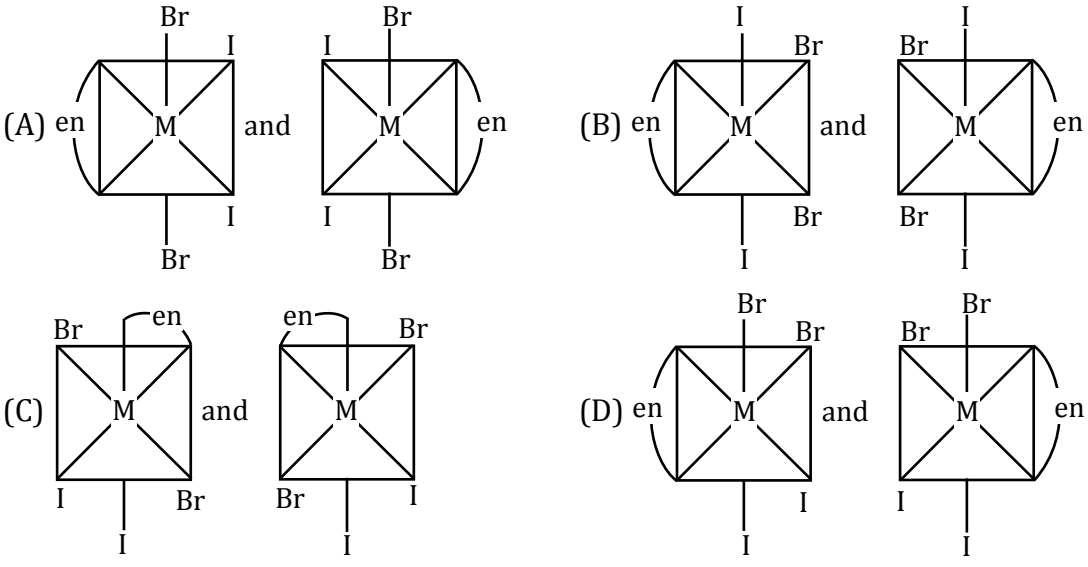
CCC209

Bankers Academy JEE

82. Find complex which have maximum number of stereoisomers -
(A) $[Ma_3b_3]$ (B) $[Ma_3b_2c]$ (C) $[Ma_2b_2c_2]$ (D) $[M(AA)a_2b_2]$

CCC222

83. The complex ion has two optical isomers. Their **CORRECT** configurations are:



CCC226

84. Which of the following is not optically active :-
(A) $[Co(en)_3]^{3\oplus}$ (B) $[Cr(ox)_3]^{3\ominus}$
(C) cis $-$ $[CoCl_2(en)_2]^{\oplus}$ (D) trans $-$ $[CoCl_2(en)_2]^{\oplus}$

CCC224

For More Join: @IITJEE_Advanced

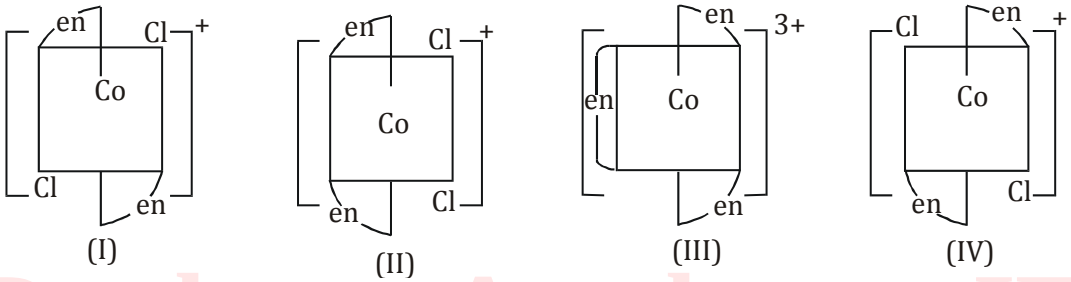
85. **Statement-1**: Complexes containing three bidentate groups such as $[Cr(ox)_3]^{3\ominus}$ and $[Co(en)_3]^{3\oplus}$ do not show optical activity.
Statement-2: Octahedral complex, $[Co(NH_3)_4Cl_2]Cl$ shows geometrical isomerism.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CCC238

86. Which of the following will show optical isomerism but not geometrical isomerism.
(A) $[Cr(Cl_3)_3(NH_3)_3]$ (B) $K_3[Fe(C_2O_4)_3]$
(C) $[CoCl_2(en)_2]^{\oplus}$ (D) $[CoBrCl(Ox)_2]^{3\ominus}$

CCC242

87. Which of the following ions are optically active ?



(A) I only (B) II only
(C) II and III (D) IV only

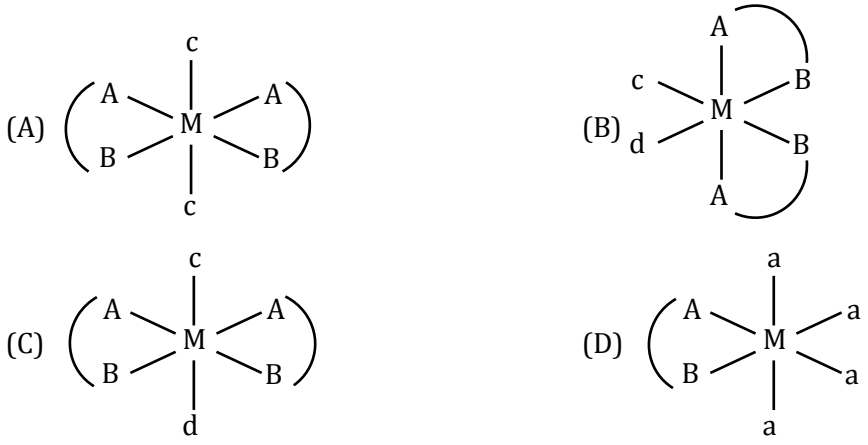
CCC223

Stereoisomerism for Octahedral Complexes for Unsymmetrical Bidentate Ligand

88. Find complex which have maximum number of stereoisomers -
(A) $[Ma_6]$ (B) $[Ma_3b_3]$
(C) [M(AB)abcd] (D) $[M(AA)a_2b_2]$

CCC225

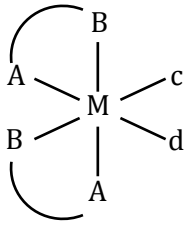
89. Which of the following octahedral complex is optically active ?
[M = Central atom, (a, b, c, d = monodentate ligands) , (AB = unsymmetrical bidentate ligand)]

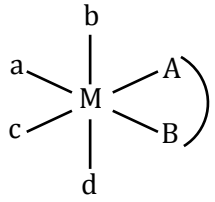


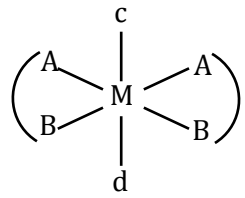
CCC227

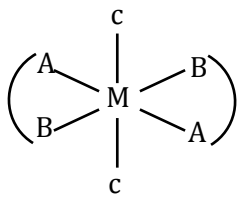
Pankers Academy JEE
For More Join: @IITJEE_Advanced

90. Select the complex which has centre of symmetry : [M = central atom, AB = unsymmetrical bidentate ligand, (a, b, c, d = monodentate ligands)]

(A) 

(B) 

(C) 

(D) 

CCC228

91. The no of geometrical isomers for the complex $[MCl_2(L)_2]$ {where M = central atom, $L = NH_2 - CH_2COO^{\ominus}$} are :

(A) 2 (B) 4 (C) 5 (D) 6

CCC230

Bonding in Coordination Compound; Valence Bond Theory; Questions of Valence Bond Theory; Limitation of Valence Bond Theory

92. Which of the following is the limitation of valence bond theory ?

(A) It does not distinguish between weak and strong ligands.

(B) It does not give quantitative interpretation of magnetic data.

(C) It does not explain the colour exhibited by coordination compounds.

(D) All of these

Bankers Academy JEE

CCC231

Spectro chemical series

93. Which complex of $Co^{2\oplus}$ will have the weakest crystal field splitting -

(A) $[CoCl_6]^{4\ominus}$ (B) $[Co(CN)_6]^{4\ominus}$ (C) $[Co(NH_3)_6]^{2\oplus}$ (D) $[Co(en)_3]^{2\oplus}$

CCC257

Crystal Field Theory for Octahedral Complexes

94. In an octahedral crystal field, the t_{2g} orbitals are

(A) Raised in energy by 0.4 Δ_O (B) Lowered in energy by 0.4 Δ_O

(C) Raised in energy by 0.6 Δ_O (D) Lowered in energy by 0.6 Δ_O

CCC266

95. Consider the following complex $[Co(NH_3)_5(CO_3)]ClO_4$ The coordination number, oxidation number, number of d-electrons and number of unpaird d-electrons on the metal are respectively

(A) 6, 3, 6, 0 (B) 7, 2, 7, 1 (C) 7, 1, 6, 4 (D) 6, 2, 7, 3

CCC232

96. Which of the following facts about the complex $[Cr(NH_3)_6]Cl_3$ is wrong ?

(A) The complex involves d^2sp^3 hybridisation and is octahedral in shape.

(B) The complex is paramagnetic.

(C) The complex is an outer orbital complex

(D) The complex gives white precipitate with silver nitrate solution.

CCC234

For More Join: @IITJEE_Advanced

97. For which of the following types of d^n configuration, the number of unpaired electrons in octahedral complexes remains same irrespective of the ligand field strength.
(A) d^3 (B) d^4 (C) d^5 (D) d^6

CCC251

98. Which of the following compound is paramagnetic –
(A) Tetracyanonickelate(II) ion (B) Tetraamminezinc(II) ion
(C) Hexaaminechromium(III) ion (D) Diamminesilver(I) ion

CCC252

Exceptions in Octahedral Complexes

99. The number of unpaired electrons calculated in $[Co(NH_3)_6]^{3\oplus}$ and $[CoF_6]^{3\ominus}$ are :
(A) 4 and 4 (B) 0 and 2 (C) 2 and 4 (D) 0 and 4

CCC264

100. $[M(C_2O_4)_3]^{x\ominus}$ is diamagnetic when $M^{n\oplus}$ ion is :
(A) $Co^{3\oplus}$ (B) $Co^{2\oplus}$ (C) $Ni^{2\oplus}$ (D) $Fe^{2\oplus}$

CCC269

101. $[NiF_6]^{2\ominus}$ is :
(A) Paramagnetic (B) high spin (C) Diamagnetic (D) tetrahedral

CCC268

102. Which of the following is correctly matched?
(A) $[Co(NH_3)_6]Cl_3$: diamagnetic, Δ_O < pairing energy
(B) $K_4[Fe(CN)_6]$: Low spin complex, colourless
(C) $[Co(H_2O)_6]SO_4$: Paramagnetic, high spin complex
(D) $[NiF_6]^{2\ominus}$ diamagnetic, Δ_O < pairing energy

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CCC220

Exceptional Cases in Cr; Mn; Fe; Co; Ni Complexes

103. Which one of the following complex is an outer orbital complex -
(A) $[Ni(H_2O)_6]^{2\oplus}$ (B) $[Fe(H_2O)_5NO]SO_4$ (C) $[Fe(NH_3)_6]^{2\oplus}$ (D) All of these

CCC256

104. The complex $[Fe(NH_3)_6]^{2\oplus}$ is :
(A) Low spin (B) High spin
(C) Diamagnetic (D) both (A) and (C)

CCC267

Crystal Field Theory for Tetrahedral Complexes

105. The geometry of $[Ni(CO)_4]$ and $[Ni(PPh_3)_2Cl_2]$ are
(A) both square planar (B) tetrahedral and square planar
(C) both tetrahedral (D) None of these

CCC233

106. Which of the following complex shows sp^3d^2 hybridization ?
(A) $[Cr(NO_2)_6]^{3\ominus}$ (B) $[Fe(CN)_6]^{4\ominus}$ (C) $[CoF_6]^{3\ominus}$ (D) $[Ni(CO)_4]$

CCC236

107. What is the hybridisation of Fe in $[Fe(CO)_4]^{2\ominus}$
(A) d^2sp^3 (B) sp^3d^2 (C) sp^3d (D) sp^3

CCC247

108. Identify tetrahedral species which has maximum magnetic moment value :
(A) $[CuCl_4]^{2\ominus}$ (B) $[CoCl_4]^{2\ominus}$ (C) $[FeCl_4]^{2\ominus}$ (D) $[AlCl_4]^{\ominus}$

CCC249

For More Join: @IITJEE_Advanced

109. Tetrahedral crystal field splitting energy Δ_t is roughly how many times of octahedral crystal field splitting energy Δ_O ?
(A) 1/2 (B) 4/9 (C) 9/4 (D) 2
CCC271

110. What CFSE of a tetrahedral complex with d^2 configuration of central atom ?
(A) – 0.6 Δ_t (B) – 1.2 Δ_t (C) – 0.8 Δ_t (D) 0
CCC276

111. How many unpaired electrons are there in d^7 tetrahedral complex ?
(A) 3 (B) 1 (C) 2 (D) 4
CCC277

112. The value of crystal field stabilisation energy for a tetrahedral complex having d^5 configuration of central metal ion is :
(A) – 0.6 Δ_t (B) – 1.2 Δ_t (C) – 0.4 Δ_t (D) Zero
CCC273

Calculation of Crystal Field Stabilisation Energy for Octahedral Complexes

113. CFSE in d^3, octahedral is :
(A) – 1.2 Δ_O (B) – 0.8 Δ_O
(C) +0.6 Δ_O (D) – 2.4 Δ_O
CCC299

114. If Δ_0 is crystal field spiliting energy and P is mean pairing energy per pair then stabilizing energy of complex $[FeF_6]^{4\ominus}$ is -
(A) $-2.4\ \Delta_0 + 3P$ (B) $-2.0\ \Delta_0 + 3P$
(C) $-1.6\ \Delta_0 + P$ (D) $-0.4\ \Delta_0$
CCC300

Parkers Academy JEE

Crystal Field Theory for Square Planar Complexes

115. The metal complex that is diamagnetic is (Atomic numer: Fe = 26; Cu = 29)
(A) $K_3[Cu(CN)_4]$ (B) $K_2[Cu(CN)_4]$ (C) $K_3[Fe(CN)_4]$ (D) $K_4[FeCl_6]$
CCC235

116. Which of the following complexes has square planar structure ?
(A) $[Ni(CO)_4]$ (B) $[Ni(CN)_4]^{2\ominus}$ (C) $[Ni(Cl)_4]^{2\ominus}$ (D) $[Zn(NH_3)_4]^{2\oplus}$
CCC237

117. Which of the following is diamagnetic and sp^3 hybridised -
(A) $[NiCl_4]^{2\ominus}$ (B) $[Ni(CN)_4]^{4\ominus}$ (C) $[Ni(CN)_4]^{2\ominus}$ (D) $[NiCl_2(PPh_3)_2]$
CCC248

118. $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2\ominus}$ do not differ in
(A) magnetic moment (B) oxidation number of Ni
(C) geometry (D) EAN
CCC250

119. The magnetic moment of $[NiX_4]^{2\ominus}$ ion is found to be zero. Then the metal of the complex ion is (X = monodentate anionic ligand).
(A) sp^3 hybridised (B) spd^2 hybridised (C) dsp^2 hybridised (D) d^2sp hybridised
CCC255

For More Join: @IITJEE_Advanced

120. One unknown complex has the spin only magnetic moment is of 1.73 BM. As per the C. F. T., complex is.
(A) d^7, Oh-field, with Strong Field Ligand (B) d^9, sq.planar-field, with Strong Field Ligand
(C) d^9, Td,field with Weak Field Ligand (D) All of these

CCC274

121. How many unpaired electrons are there in square planar complex of $Co^{2\oplus}$?
(A) 1 (B) 2 (C) 3 (D) 4

CCC279

122. The number of unpaired electron in $[AuCl_4]^{\ominus}$ is :
(A) 0 (B) 1 (C) 2 (D) 4

CCC280

123. $[PdCl_4]^{2\ominus}$ is :
(A) Tetrahedral, 2 unpaired e^-s. (B) Square planar, paramagnetic
(C) Square planar, diamagnetic (D) Tetra hedral, diamagnetic

CCC281

124. Which of the following is a correct statement?
(A) $\left[Co(NH_3)_6\right]^{3\oplus}$ is paramagnetic while $\left[CoF_6\right]^{3\ominus}$ is diamagnetic
(B) $\left[Pt(NH_3)(NH_2OH)(Py)(NO_2)\right]NO_3$ has square planar shape
(C) $Ni(CO)_4$ possesses square planar geometry while $\left[Ni(CN)_4\right]^{2\ominus}$ is tetrahedral.
(D) IUPAC name for $\left[Co(NH_3)_4Cl(ONO)\right]Cl$ is tetraamminechloronitrocobalt (II) chloride.

CCC311

125. The compound Na_2IrCl_6 reacts with triphenylphosphine in diethyleneglycol in an atmosphere of CO to give $[IrCl(CO)(PPh_3)_2]$, known as 'Vaska's compound'. (Atomic number of Ir = 77) Which of the following statements is correct?
(A) The IUPAC name of the complex is carbonylchloridobis (triphenylphosphine) iridium(III)
(B) The hybridisation of the metal ion is sp^3
(C) The magnetic moment (spin only) of the complex is zero
(D) The complex shows geometrical as well as ionization isomerism

CCC289

Rankers Academy JEE

Factors Affecting Crystal Field Stabilisation Energy

126. Crystal field stabilisation energy of a complex is affected by :
(A) Ligand strength (B) Oxidation Number of central atom
(C) Coordination number (D) All of these

CCC296

127. Δ_O is maximum for which of the following given complex?
(A) $[Co(H_2O)_3F_3]$ (B) $[Co(NH_3)_6]^{3\oplus}$
(C) $[Co(H_2O)_6]^{3\oplus}$ (D) $[Co(NH_3)_4]^{3\oplus}$

CCC297

128. Select the correct general order of comparison :
(A) $\Delta_t > \Delta_{sp} > \Delta_O$ (B) $\Delta_{sp} > \Delta_O > \Delta_t$
(C) $\Delta_{sp} > \Delta_t > \Delta_O$ (D) $\Delta_O > \Delta_{sp} > \Delta_t$

CCC298

129. Among the following species the one which causes the highest CFSE, Δ_O as a ligand is :-
(A) $CN^{\ominus}$ (B) NH_3 (C) CO (D) $F^{\ominus}$

CCC301

For More Join: @IITJEE_Advanced

Colour in Transition Metal Coordination Compound Using d - d Transition; Charge Transfer Spectrum; Complementary Colour Wheel;

130. $[Sc(H_2O)_6)]^{3\oplus}$ ion is :
(A) colourless and diamagnetic
(B) coloured and octahedral
(C) colourless and paramagnetic
(D) coloured and paramagnetic

CCC291

131. An ion $M^{2\oplus}$, forms the complexes $[M(H_2O)_6]^{2\oplus}$, $[M(en)_3]^{2\oplus}$ and $[MBr_6]^{4\ominus}$, match the complex with the appropriate colour.
(A) Green, blue and red
(B) Blue, red and green
(C) Green, red and blue
(D) Red, blue and green

CCC302

132. If $\lambda_{absorbed}$ for d-d transition is in order $[Ti(X)_6]^{3\oplus} > [Ti(Y)_6]^{3\oplus} > [Ti(Z)_6]^{3\oplus}$.
Select correct order of strength of ligands (X, Y, Z are monodentate ligand)-
(A) Z > Y > X
(B) X > Y > Z
(C) Z > X > Y
(D) Not predictable

CCC303

133. Which of the following complex is coloured and diamagnetic –
(A) $MnO_4^{2\ominus}$
(B) $[Ni(H_2O)_6]^{2\oplus}$
(C) $[CrCl_6]^{3\ominus}$
(D) $CrO_4^{2\ominus}$

CCC304

134. Which of the following compound is not coloured ?
(A) $Na_2[CuCl_4]$ (B) $Na_2[CdCl_4]$ (C) $K_4[Fe(CN)_6]$ (D) $K_3[Fe(CN)_6]$

CCC307

Rankers Academy JEE

135. Given the following data about the absorption maxima of several complex ions, what is the order of Δ_0 for these ions.

Compound	λ_{max}
I. $[CrCl_6]^{3\ominus}$	758
II. $[Cr(NH_3)_6]^{3\oplus}$	465
III. $[Cr(H_2O)_6]^{3\oplus}$	694

(A) Δ_0 (I) < Δ_0 (II) < Δ_0 (III)
(B) Δ_0 (II) < Δ_0 (III) < Δ_0 (I)
(C) Δ_0 (I) < Δ_0 (III) < Δ_0 (II)
(D) Δ_0 (III) < Δ_0 (II) < Δ_0 (I)

CCC308

Synergic Bonding

136. Statement-1 : In Zeise's salt C – C bond order regarding ethylene molecule is less than two.
Statement-2 : Ethylene accepts electron pair from filled d-orbital of $Pt^{2\oplus}$ into its vacant bonding M.O.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CCC120

137. Which of the following complex has highest M – C bond order ?
(A) $[Mn(CO)_6]^{\oplus}$ (B) $[V(CO)_6]^{\ominus}$ (C) $[Cr(CO)_6]$ (D) $[Ti(CO)_6]^{2\ominus}$

CCC135

For More Join: @IITJEE_Advanced

138. In the isoelectronic series of metal carbonyl, the C–O bond strength is expected to increase in the order.
(A) $[Mn(CO)_6]^{\oplus} < [Cr(CO)_6] < [V(CO)_6]^{\ominus}$
(B) $[V(CO)_6]^{\ominus} < [Cr(CO)_6] < [Mn(CO)_6]^{\oplus}$
(C) $[V(CO)_6]^{\ominus} < [Mn(CO)_6]^{\oplus} < [Cr(CO)_6]$
(D) $[Cr(CO)_6] < [Mn(CO)_6]^{\oplus} < [V(CO)_6]^{\ominus}$

CCC136

139. Which of the following order is correct for the IR vibrational frequency of CO.
(A) $[Fe(CO)_4]^{2\ominus} < [Co(CO)_4]^{\ominus} < [Ni(CO)_4]$
(B) $[Fe(CO)_4]^{2\ominus} > [Co(CO)_4]^{\ominus} > [Ni(CO)_4]$
(C) $[Fe(CO)_4]^{2\ominus} > [Co(CO)_4]^{\ominus} < [Ni(CO)_4]$
(D) $[Fe(CO)_4]^{2\ominus} < [Co(CO)_4]^{\ominus} > [Ni(CO)_4]$

CCC137

140. Which of the following complex has highest C–O bond length ?
(A) $[Ni(CO)_3(PF_3)]$ (B) $[Ni(CO)_4]$
(C) $[Ni(CO)_3(PMe_3)]$ (D) None of these

CCC138

141. Metal-ligand bond order is more than 1 in which of the following complex species ?
(A) $[Co(NH_3)_6]Cl_3$ (B) $[Cr(CN_6)]^{3\ominus}$ (C) $[V(CO)_6]^{\ominus}$ (D) $[Sc(H_2O)_6]Cl_3$

CCC139

142. Select the incorrect statement about metal carbonyl complex compounds :
(A) Metal carbon bond in metal carbonyls possess both σ and π character
(B) Due to synergic bonding metal carbon bond becomes weak
(C) Due to synergic bonding carbon oxygen bond strength decreases
(D) In metal carbonyls extent of synergic bonding will increase with increase in negative charge on central metal ion

Bansal's Academy JEE

CCC140

Importance and Applications of Coordination Compounds

143. Which one of the following coordination compounds is used to inhibit the growth of tumours ?
(A) Trans-platin (B) EDTA complex of calcium
(C) $[(Ph_3P)_3RhCl]$ (D) Cis-platin

CCC283

144. The compound used in the treatment of lead poisoning is :
(A) D-penicillamine (B) desferrioxime B
(C) Cis-platin (D) EDTA

CCC284

145. Coordination compounds have great importane in biological system. In this context which of the following statements is incorrect.
(A) Cyanocobalamin is B_{12} and contains cobalt
(B) Haemoglobin is the red pigment of blood and contains iron
(C) Chlorophylls are green pigments in plants and contain calcium
(D) Carboxypeptidase – A is an enzyme and contains zinc.

CCC285

146. Which of the following is not considered as an organometallic compound ?
(A) Ferrocene (B) Cis-platin (C) Ziese's salt (D) Grignard reagent

CCC121

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Introduction of Coordination Compounds

1. Which is TRUE about metal complex ?
(A) Central metal acts as lewis acid
(B) Ligands acts lewis base
(C) It will give test of all individual entities in aqueous solution
(D) The coordinate bonds are present in the metal complex

CCC108

Types of Complex: Homoleptic and Heteroleptic Complex

2. Which of the following complexes is/are an example of homoleptic complex ?
(A) $[PtCl_4]^{2\ominus}$ (B) $K_3[Fe(CN)_6]$
(C) $[Cr(NH_3)_3Cl_3]$ (D) $[CoBr_2(en)_2]^{\oplus}$

CCC109

3. Which of the following complexes is/are an example of heteroleptic complex ?
(A) $[Co(NO_2)_3(NH_3)_3]$ (B) $K_3[CrCl_6]$
(C) $[Co(NH_3)_3Cl_3]$ (D) $[Cr(C_2O_4)_3]^{3\ominus}$

CCC110

Types of Ligands: On the basis of Denticity; Ligand as Polydentate Ligand; Chelating Ligands; Ambidentate Ligands; Flexidentate Ligands;

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4. Which of the following bidentate ligand(s) has similar donor atoms :
(A) py (B) bn (C) en (D) gly^-

CCC129

Classical and Non Classical Ligands

5. Which of the following chemical species can act as non-classical ligand(s)
(A) CO (B) C_2H_4 (C) $NO^{\oplus}$ (D) PR_3

CCC130

Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

6. Which of the following complex is following sidwick EAN rule.
(A) $[Ni(CO)_4]$ (B) $K_2[Ni(CN)_4]$ (C) $K_2[HgI_4]$ (D) $[Ag(NH_3)_2]Cl$

CCC168

7. Which of the following pairs of compounds has the same EAN value but does not obey Sidgwick EAN rule?
(A) $Fe(CO)_5$, $Ni(CO)_4$ (B) $[Fe(NH_3)_6]^{2+}$, $[Cr(C_2O_4)_3]^{3-}$
(C) $[Fe(H_2O)_5NO]^{2+}$, $[Mn(CO)_6]$ (D) $[Ti(CO)_6]$, $[Mn(C_2O_4)_3]^{3-}$

CCC324

Stereoisomerism for Square Planar Complexes

8. Select which Square planar complex (es) cannot show optical isomerism : -
(A) $[Pt(bn)_2]^{2\oplus}$ (B) $[Pt(en)_2]^{2\oplus}$ (C) $[PtCl_2(NH_3)_2]^0$ (D) $[Pt(gly)_2]^0$

CCC212

For More Join: @IITJEE_Advanced

Stereoisomerism for Octahedral Complexes for Monodentate Ligand

9. Which of the following exhibit geometrical isomerism (M stands for a metal, and a and b are achiral ligands) ?

(A) Ma_2b_2 (Sq. Pl.) (B) Ma_4b_2 (C) Ma_5b (D) Ma_6

CCC240

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

10. Which of the following will have two stereoisomeric forms?

(A) $[Cr(NO_3)_3(NH_3)_3]$ (B) $K_3[Fe(C_2O_4)_3]$

(C) $[CoCl_2(en)_2]^{\oplus}$ (D) $[CoBrCl(Ox)_2]^{3\ominus}$

CCC241

Exceptions in Octahedral Complexes

11. $[MnCl_6]^{3\ominus}$, $[FeF_6]^{3\ominus}$ and $[CoF_6]^{3\ominus}$ complexes are -

(A) Outerorbital

(B) Having sp^3d^2 hybridisation

(C) Paramagnetic in nature

(D) Having four, five and four unpaired electrons respectively

CCC262

Exceptional Cases in Cr; Mn; Fe; Co; Ni Complexes

12. In case of $[Mn(NH_3)_6]^{2+}$

(A) relation in between Δ_0 and P is $\Delta_0 < P$.

(B) two unpaired electrons are present in axial d-orbital of Mn^{2+}.

(C) hybridization of central metal atom of complex is d^2sp^3.

(D) d-orbitals involved in hybridization of central atom are d_{x^2} and d_{xy}.

Rankers Academy JEE

CCC322

Crystal Field Theory for Tetrahedral Complexes

13. Which of the following ion is/are diamagnetic and non planar ?

(A) $[Ni(CN)_4]^{2\ominus}$ (B) $MnO_4^{\ominus}$

(C) $[Cu(NH_3)_4]^{2\oplus}$ (D) $CrO_4^{2\ominus}$

CCC315

Crystal Field Theory for Square Planar Complexes

14. Select pair(s) in which given complex is isostructural with given xenon fluoride.

(A) XeF_2 ; $[AgCl_2]^{\ominus}$ (B) XeF_4 ; $[NiCl_4]^{2\ominus}$ (C) XeF_4 ; $[PtCl_4]^{2\ominus}$ (D) XeF_6 ; $[PtCl_4]^{2\ominus}$

CCC263

15. Which of the following statement(s) is/are correct?

(A) $Ni(CO)_4$ is diamagnetic and sp^3 hybridized.

(B) Complex $[Cu(CN)_4]^{3-}$ is paramagnetic and Cu^{2+} ion is dsp^2 hybridized.

(C) $[MnBr_4]^{2-}$ ion is paramagnetic and sp^3 hybridized.

(D) Vaska's catalyst, that is, $[IrCl(CO)(PPh_3)_2]$ is diamagnetic and Ir^+ ion is dsp^2 hybridized.

CCC321

16. Which of the following complexes is/are square planar?

(A) $[Ag\ F_4]^-$ (B) $[AuCl_4]^-$ (C) $[NiCl_2(PPh_3)_2]$ (D) $[NiCl_2(Pme_3)_2]$

CCC323

For More Join: @IITJEE_Advanced

17. In which of the following complex(s) spin only magnetic moment is independent , from the nature of ligand. (L = monodented ligand) -

(A) $[NiL_4]^{2\oplus}$ (B) $[NiL_6]^{2\oplus}$ (C) $[FeL_6]^{3\oplus}$ (D) $[CrL_6]^{3\oplus}$

CCC314

Factors Affecting Crystal Field Stabilisation Energy

18. Select the correct stability order for the following complexes.

(A) $[Cu(trien)]^{2\oplus} > [Cu(en)_2]^{2\oplus} > [Cu(NH_3)_4]^{2\oplus}$

(B) $[Ir(H_2O)_6]^{3\oplus} > [Rh(H_2O)_6]^{3\oplus} > [Co(H_2O)_6]^{3\oplus}$

(C) $[Cr(NH_3)_6]^{3\oplus} > [Cr(NH_3)_6]^{2\oplus} > [Cr(NH_3)_6]^{\oplus}$

(D) $[Cu(CO)_4]^{2\oplus} > [Cu(H_2O)_4]^{2\oplus} > [Cu(NH_3)_4]^{2\oplus}$

CCC313

METRIX MATCH TYPE QUESTIONS

Types of Ligands: On the basis of Denticity; Ligand as Polydentate Ligand; Chelating Ligands; Ambidentate Ligands; Flexidentate Ligands;

19. Match each coordination compound in List-I with co-ordination number of central metal/ion from List-II and select the correct answer using the code given below the lists.

List-I	**List-II**
(P) $[Co(en)_3]^{3\oplus}$	(1) 6
(Q) $[Ca(EDTA)]^{2\ominus}$	(2) 4
(R) $[Ni(CO)_4]$	(3) 2
(S) $[Ag(NH_3)_2]Cl$	(4) 5

(A) $P \rightarrow (1), Q \rightarrow (3), R \rightarrow (2), S \rightarrow (3)$ (B) $P \rightarrow (1), Q \rightarrow (1), R \rightarrow (2), S \rightarrow (3)$

(C) $P \rightarrow (2), Q \rightarrow (1), R \rightarrow (2), S \rightarrow (2)$ (D) $P \rightarrow (2), Q \rightarrow (2), R \rightarrow (3), S \rightarrow (2)$

CCC127

Rankers Academy JEE

20. Match the List with multiple options

List-I	**List-II**
(P) Ethane 1, 2 diamine	(1) Chelating agent
(Q) $SCN^{\ominus}$	(2) Ambidentate
(R) $CO_3^{2\ominus}$	(3) Flexidentate
(S) $NO_2^{\ominus}$	(4) Tridentate

Code :

	P	**Q**	**R**	**S**
(A)	1	2	3	4
(C)	4	1	3	2

	P	**Q**	**R**	**S**
(B)	1	2	3	2
(D)	4	1	2	4

CCC128

Werner Theory; Electrical and Molar Conductivity

21. Match List I with List II and select the corect answer using the codes given below the lists :-

List I (Equivalent (conductance)	**List II (Formula)**
A. 229	i. $[Pt(NH_3)_5\ Cl]Cl_3$
B. 97	ii. $[Pt(NH_3)_3Cl_3]Cl$
C. 404	iii. $[Pt(NH_3)_4Cl_2]Cl_2$
D. 523	iv. $[Pt(NH_3)_6]Cl_4$

(A) A→i; B→iii; C→ii; D→iv (B) A→i; B→iv; C→ iii; D→ii

(C) A→i; B→iii; C→ iv; D→ii (D) A→iii; B→ii; C→i; D→iv

CCC152

For More Join: @IITJEE_Advanced

Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

22. Match the complexes in column-I with the EAN of central atom in column-II:

Column-I	**Column-II**
(A) $[Fe(CO)_4]^{2\ominus}$	(P) 34
(B) $[Co(NH_3)_5Cl]Cl_2$	(Q) 35
(C) $K_2[Ni(CN)_4]$	(R) 36
(D) $[Cu(NH_3)_4]^{2\oplus}$	(S) 37

(A) A-(P), B-(R), C-(P), D-(S) (B) A-(R), B-(R), C-(P), D-(Q)
(C) A-(P), B-(S), C-(R), D-(Q) (D) A-(R), B-(S), C-(P), D-(Q)

CCC163

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

23. Match the complexes in column I with their stereoproperties is column II

Column I	**Column II**
(A) $[CoCl_3(NH_3)_3]$	(P) Has a facial isomer
(B) $[Cr\ (ox)_3]^{3\ominus}$	(Q) Cis form is optically active
(C) $[CrCl_2(ox)_2]$	(R) Trans form is optically inactive
(D) $[RhCl_3(Py)_3]$	(S) Has a meridional form
	(T) Two optically active isomer

(A) A–P,R B–T, C–Q,S,T D–P,Q (B) A–P,S B–T, C–Q,S,T D–P,S
(C) A–P,S B–T, C–Q,R,T D–P,S (D) A–P,R B–T, C–Q,R,T D–P,Q

CCC239

Crystal Field Theory for Octahedral Complexes

Konkers Academy JEE

24.

Column - I	**Column -II**
(A) $Na_2\ [Fe(CN)_5\ NO]$	(P) $\mu = 0$ B.M.
(B) $[Fe(H_2O)_5\ NO]\ SO_4$	(Q) octahedral
(C) $[Ag(CN)_2]^{\ominus}$	(R) $\mu = \sqrt{15}$ B.M.
(D) $K_4[Fe(CN)_6]$	(S) $NO^{\oplus}$ ligand

(A) (A–P,Q,R B–Q,R,T C–P D–P,R) (B) (A–P,Q,S B–Q,R,S C–P D–P,Q)
(C) (A–P,Q,S B–Q,R,T C–Q D–P,Q) (D) (A–P,Q,R B–Q,R,S C–P D–P,S)

CCC258

25. Match List–I (Complex ions) with List–II (Number of Unpaired Electrons) and select the correct answer using the codes given below the lists :-

List –I (Complex ions)	**List II (Number of Unpaired Electrons)**
A. $[CrF_6]^{4\ominus}$	i. One
B. $[MnF_6]^{4\ominus}$	ii. Two
C. $[Cr(CN)_6]^{4\ominus}$	iii. Three
D. $[Mn(CN)_6]^{4\ominus}$	iv. Four
	v. Five

Code :

	A	**B**	**C**	**D**
(A)	iv	i	ii	v
(C)	iv	v	ii	I

	A	**B**	**C**	**D**
(B)	ii	v	iii	i
(D)	ii	i	iii	v

CCC272

For More Join: @IITJEE_Advanced

Crystal Field Theory for Square Planar Complexes

Answer Q.26, Q.27 and Q.28 by appropriately matching the information given in the three columns of the following table.

For the following complexes in Column-I, match the correct properties with Column-II and Column-III according to the questions asked.

Column-I	Column-II Hybridisation	Column-III Magnetic moment value in B.M.
(I) $[Co(H_2O)_3F_3]$	(i) sp^3	(p) 2.8
(II) $[NiF_6]^{2\ominus}$	(ii) dsp^2	(q) 5.92
(III) $Na_2[NiCl_4]$	(iii) sp^3d^2	(r) 0
(IV) $K_2[Ni(CN)_4]$	(iv) d^2sp^3	(s) 4.9

26. Which is the only correct combination?

(A) I, (iv), (r) (B) II, (i), (q) (C) IV, (ii), (p) (D) IV, (ii), (r)

CCC259

27. Which is the only incorrect combination ?

(A) IV, (ii), (r) (B) II, (iii), (p) (C) II, (iv), (r) (D) III, (i), (p)

CCC260

28. Which is the only correct combination for $[Co(H_2O)_3F_3]$?

(A) (iv), (r) (B) (iii), (p) (C) (iii), (s) (D) (iv), (q)

CCC261

Bankers Academy JEE

Synergic Bonding

29. **Match the list :**

List-I **[Species]**	**List-II** **[Stretching frequency of CO ($cm^{1\ominus}$)]**
(P) CO	(1) 2143
(Q) $Cr(CO)_6$	(2) 2000
(R) $[V(CO)_6]^{\ominus}$	(3) 1860
(S) $[Mn(CO)_6]^{\oplus}$	(4) 2090

(A) P→1; Q→2; R→3; S→4 (B) P→2; Q→1; R→3; S→4

(C) P→3; Q→1; R→2; S→4 (D) P→4; Q→1; R→3; S→2

CCC151

Importance and Applications of Coordination Compounds

30. Match the List-I with List-II :

List-I	**List-II**
(P) Ferrocene	(1) Iron present
(Q) $Mn_2(CO)_{10}$	(2) Cobalt
(R) Vitamine B_{12}	(3) Metal-Metal bonding
(S) Haemoglobin	(4) Sandwich structure

Code :

	P	Q	R	S
(A)	4	3	1	2
(B)	1	3	1	2
(C)	1,4	3	2	1
(D)	1	3	4	2

CCC309

For More Join: @IITJEE_Advanced

Table for Question 31 to 33

Complex		Oxidation state of central metal atom or ion		Ligand (s) present	
(1)	Ferrocene	(P)	0	(i)	$C_5H_5^{\ominus}$
(2)	Potassium ferrocynide	(Q)	+1	(ii)	$NO^{\oplus}$
(3)	Sodium nitroprusside	(R)	+2	(iii)	$CN^{\ominus}$
(4)	Zeise's salt	(S)	+3	(iv)	$Cl^{\ominus}$

31. Which of the following matching is INCORRECT : -
(A) (1),(R),(i) (B) (2), (R), (iii)
(C) (3), (R), (ii) and (iii) (D) (4), (S), (iv)

CCC189

32. Select INCORRECT combination for complex given : -
(A) Complex having platinum = R, (iv)
(B) Neutral / uncharged complex = R, (i)
(C) Complex which have η^2 – bonded ligand = R, (i)
(D) Sandwich complex = R, (i)

CCC190

33. Which of the following complex having π-acid ligand (s) : -
(A) 1,2,3,4 (B) 1, 2, 3 (C) 1, 2 (D) 1, 4

CCC191

Rankers Academy JEE

COMPREHENSION TYPE QUESTIONS

Stereoisomerism for Octahedral Complexes for Monodentate Ligand

Paragraph for Question No. *34 to 36*

Coordination number of a central atom is its inherent property and has no role on its isomerism.
$[CoCl_2(NH_3)_4]^+ + Cl^- \rightarrow [CoCl_3(NH_3)_3] + NH_3$

34. If in the above reaction two isomers of the product are obtained, which is true for the initial (reactant) complex?
(A) compound is in cis form.
(B) compound is in trans form.
(C) compound is in both (cis and trans) forms.
(D) cannot be predicted.

CCC328

35. When 0.1 mol of $CoCl_3(NH_3)_5$ is treated with excess of $AgNO_3$, 0.2 mol of AgCl are obtained. The conductivity of solution will correspond to (cation : anion)
(A) 1 : 3 electrolyte (B) 1 : 2 electrolyte
(C) 1 : 1 electrolyte (D) 3 : 1 electrolyte

CCC329

36. Due to the presence of ambidentate ligands coordination compounds show isomerism, palladium complexes of the type $[Pd(C_6H_5)_2(SCN)_2]$ and $[Pd(C_6H_5)_2(NCS)_2]$ are
(A) linkage isomers. (B) coordination isomers.
(C) ionisation isomers. (D) geometrical isomers.

CCC330

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Introduction of Coordination Compounds

1. Find the number of correct statement(s) is/are :-

(I) Both double salts as well as complexes are formed by the combination of two or more stable compounds in stoichiometric ratio

(II) double salts such as carnallite, $KCl.MgCl_2.6H_2O$, potash alum, $KAl(SO_4)_2.12H_2O$, etc. dissociate into simple ions completely when dissolved in water

(III) complex ions such as $[Fe(CN)_6]^{4\ominus}$ or $K_4[Fe(CN)_6]$, do not dissociate into $Fe^{2\oplus}$ and $CN^{\ominus}$ ions

(IV) Mohr's salt, $FeSO_4.(NH_4)_2SO_4.6H_2O$ does not dissociate into simple ions completely when dissolved in water

(V) $[CoCl_2(NH_3)_4]Cl$ complex dissociate into $Co^{3\oplus}$ and $Cl^{\ominus}$ ions

CCC111

Types of Ligands: On the basis of Denticity; Ligand as Polydentate Ligand; Chelating Ligands; Ambidentate Ligands; Flexidentate Ligands;

2. What is the sum of coordination number of the central metal ions in the following complexes

(i) $[Cu(NH_3)_4]^{2\oplus}$ (ii) $[Fe(C_2O_4)_3]^{3\ominus}$ (iii) $[Pt\,Cl_2(en)_2]^{2\oplus}$

(iv) $[Mo(CN)_6]^{4\ominus}$ (v) $[Fe(EDTA)]^{\ominus}$ (vi) $[Pd\,I_2(ONO)_2(H_2O)_2]$

CCC112

3. Find the total number of 5-membered rings present in $[Co(EDTA)]^{\ominus}$.

CCC131

4. Find the number of complex(es) which have at least one five member chelate ring formed by two carbon atoms, two nitrogen atoms and one central metal.

(I) $[Co\,(EDTA)]^{\ominus}$ (II) $[Co(en)_3]^{3\oplus}$ (III) $[Co\,(Gly)_3]^{0}$ (IV) $[Co\,(bipy)_3]^{3\oplus}$

(V) $[Co(oxalate)_3]^{3\ominus}$ (VI) $[Co\,(dien)(NH_3)_3]^{3\oplus}$

CCC132

Classical and Non Classical Ligands

5. Find the number of ligands which behave predominantly as π-donor ligand (as well as non-classical ligand)

$C_2H_4, C_6H_6, NO^{\oplus}, NO, CO, NH_3, NH_2^{\ominus}, H_2O, OH^{\ominus}$

CCC133

Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

6. Determine the effective atomic number of the metal atom in $K_3[Cr(C_2O_4)_3]$

CCC170

7. Find the EAN value of central atom of $[Fe(\pi-C_4H_4)(CO)_3]$

CCC171

For More Join: @IITJEE_Advanced

Structural Isomerism

8. The number of coordination isomers possible for:

(a) $[Pt(NH_3)_4\ Cl][PtCl_4]$ is _____.

(b) $[Pt(NH_3)_4\ Cl_2][Pt(CN)_4]$ is _____.

(c) $[Fe(NH_3)_6]^{3+}\ [Cr(C_2O_4)_3]^{3-}$ is _____.

(d) $[Pt(NH_3)_4]^{2+}[CuCl_4]^{2-}$ is _____.

(e) $[Pt(NH_3)_3\ Cl]_2\ [Pt(SCN)_4]$ is _____.

CCC331

Stereoisomerism for Square Planar Complexes

9. Total number of geometrical isomers for the complex $[RhCl(CO)(PPh_3)(NH_3)]$ is.

CCC213

10. Write all the geometrical isomers of $[Pt(NH_3)(Br)(Cl)(py)]$ and how many of these will exhibit optical isomers ?

CCC214

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

11. Find the number of optically active isomers for $[Pd(en)_2(NH_3)(H_2O)]^{4\oplus}$ cation.

CCC244

12. Number of possible Geometrical Isomers of $[PtFClBrI(H_2O)(NH_3)]$

CCC246

13. Determine total number of optically active isomers of $[Co(en)_2Cl_2]Cl$.

CCC245

Rankers Academy JEE

Crystal Field Theory for Octahedral Complexes

14. Find the maximum number of atoms lying in one plane for $[Cr(CN)_6]^{3\ominus}$

CCC243

Exceptions in Octahedral Complexes

15. The number of unpaired electrons in the t_{2g} set of d orbitals in the complex $[Co(H_2O)_3F_3]$ is _____.

CCC333

Calculation of Crystal Field Stabilisation Energy for Octahedral Complexes

16. If crystal field stablization energy of $[ML_6]^{n\oplus}$ is $-0.8\ \Delta_O$.
Find minimum number of electrons in t_{2g} orbitals of metal ion ?

CCC294

Crystal Field Theory for Square Planar Complexes

17. Number of unpaired electrons in $K_3[Fe(CN)_6]$, $K_2[NiCl_4]$, $K_2[CoCl_4]$, $Na_2[Ni(CN)_4]$ are x, y, z and w respectively find x + y + z + w = ?

CCC293

18. Total number of complex compound which have low spin as well as Diamagnetic.
$[Fe(CN)_6]^{3\ominus}$, $[Co(H_2O)_6]^{3\oplus}$, $[NiF_6]^{2\ominus}$, $[Cr(H_2O)_6]^{3\oplus}$, $[Fe(CN)_6]^{4\ominus}$, $[Co(C_2O_4)_3]^{3\ominus}$, $[Ni(en)_3]^{2\oplus}$, $[PtCl_4]^{2\ominus}$

CCC295

For More Join: @IITJEE_Advanced

Factors Affecting Crystal Field Stabilisation Energy

19. Suppose we replace all the fluoride and water ligands from $[CoF_3(H_2O)_3]$ by oxalate ligand without changing the oxidation state and coordination number of central metal atom or ion. Predict the number of parameters among following which increase in the newly formed complex.
(I) Δ_0
(II) Number of t_{2g} electrons
(III) Number of e_g electrons
(IV) Number of stereoisomers
(V) Number of geometrical isomers
(VI) EAN value

CCC332

20. Find the number of complexes in which stability constant value is greater than the stability constant value of $[FeF_6]^{3\ominus}$.
$[Fe(CN)_6]^{3\ominus}$, $[Fe(ox)_3]^{3\ominus}$, $[Fe(NH_3)_6]^{3\oplus}$,
$[Fe(H_2O)_6]^{3\oplus}$, $[Fe(en)_3]^{3\oplus}$

CCC334

Synergic Bonding

21. Select complex in which metal–carbon linkage(s) is / are present :
$\left[Fe(EDTA)\right]^{\ominus}$, $K[PtCl_3(\pi\text{–}C_2H_4)]$, $[Cr(acac)_3]°$, $[Fe(CO)_5]$, $[Co(C_2H_5O)_6]^{3\ominus}$, $[Ni(dmg)_2]$

CCC153

Rankers Academy JEE

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Werner Theory; Electrical and Molar Conductivity

22. The volume (in mL) of 0.125 M $AgNO_3$ required to quantitatively precipitate chloride ions in 0.3 g of $[Co(NH_3)_6]Cl_3$ is _____.
$^M[Co(NH_3)_6]Cl_3 = 267.46$ g/mol
$^MAgNO_3 = 169.87$ g/mol

CCC336

Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

23. Consider a metal complex ion $[M(CO)_5]$. If the EAN (effective atomic number) of metal in complex is 36, then find out the following ratio for central metal atom (M) in the isolated state

$$\frac{\text{number of d electrons}}{\text{sum of number of s \& p electrons}}$$

CCC337

For More Join: @IITJEE_Advanced

Stereoisomerism for Square Planar Complexes

24. Consider the given complex ion $\left[Pt(bn)_2\right]^{2\oplus}$, where bn stands for butylene diamine. Find the value of $\frac{n}{2}$ where n is the total number of stereoisomers.

CCC338

Stereoisomerism for Octahedral Complexes for Monodentate Ligand

25. What is the number of total stereoisomers for the complex of type $[Ma_2b_2cd]$.

CCC339

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

26. Find out the ratio of optically active to optically inactive compounds among the following

$[Be(bcac)_2]$; $[CoBrCl(NO_2)_4]^{3\ominus}$

$[Zn(gly)_2]$; $[Pt(en)Cl_2]$;

$[CoF_3(H_2O)_3]$; $[Co(EDTA)]^{\ominus}$

$[Pt(en)_3]^{4\oplus}$

CCC340

Crystal Field Theory for Octahedral Complexes

27. For the $t_{2g}^6 eg^2$ system, the value of magnetic moment (μ) is :-

CCC341

28. For the co-ordination compound, $[Co(NH_3)_4Cl_2]Br$

Total number of axial d-orbitals used for hybridisation in central atom = x

Total number of stereoisomers = p

Maximum number of atoms that may lie in same plane = q

The value of $\left(\frac{q}{p.x}\right)$ is-

Rankers Academy JEE

CCC342

Calculation of Crystal Field Stabilisation Energy for Octahedral Complexes

29. For an octahedral complex of $Fe^{3\oplus}$ the value of C.F.S.E. (Crystal field stabilization energy) is "$x\Delta_0$" if magnetic moment of complex is $\sqrt{35}$ B.M. Find out the value of |x|. (Δ_0 is crystal field splitting energy in octahedral field)

CCC343

Crystal Field Theory for Square Planar Complexes

30. Consider the following complex compounds.

$[CoCl_2Br_2]^{2\ominus}$, $[Ni(en)_3]^{2\oplus}$, $[Ti(H_2O)_6]^{3\oplus}$, $[Fe(NH_3)_6]^{2\oplus}$, $[Co(H_2O)_3F_3]$, $K_3[Fe(CN)_6]$

If number of low spin complexes is "X" and number of high spin complexes is "Y" then $\frac{X}{Y}$ is _____.

CCC344

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. The complex of Ni^{2+} ion and dimethyl glyoxime contains _____ number of Hydrogen (H) atoms.
[JEE (Main) 2025]
CCC382

2. The set which does not have ambidentate ligand(s) is
[JEE (Main) 2023]

(1) $C_2O_4^{2\ominus}$, ethylene diammine, H_2O
(2) $EDTA^{4\ominus}$, $NCS^{\ominus}$, $C_2O_4^{2\ominus}$
(3) $NO_2^{\ominus}$, $C_2O_4^{2\ominus}$, $EDTA^{4\ominus}$
(4) $C_2O_4^{2\ominus}$, $NO_2^{\ominus}$, $NCS^{\ominus}$

CCC316

3. One mole of the octahedral complex compound $Co(NH_3)_5Cl_3$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with excess of $AgNO_3$ solution to yield two moles of $AgCl_{(s)}$. The structure of the complex is :
[JEE (Main) 2025]

(1) $[Co(NH_3)_5Cl]Cl_2$
(2) $[Co(NH_3)_4Cl].Cl_2.NH_3$
(3) $[Co(NH_3)_4Cl_2]Cl.NH_3$
(4) $[Co(NH_3)_3Cl_3].2NH_3$

CCC383

4. On treatment of 100 mL of 0.1 M solution of $CoCl_3.6H_2O$ with excess $AgNO_3$; 1.2×10^{22} ions are precipitated. The complex is :-
[JEE (Main) 2017]

(1) $[Co(H_2O)_4Cl_2]Cl.2H_2O$
(2) $[Co(H_2O)_3Cl_3].3H_2O$
(3) $[Co(H_2O)_6]Cl_3$
(4) $[Co(H_2O)_5Cl]Cl_2.H_2O$

CCC037

5. The correct IUPAC name of $[PtBr_2(PMe_3)_2]$ is:
[JEE (Main) 2024]

(1) bis(trimethylphosphine)dibromoplatinum(II)
(2) bis[bromo(trimethylphosphine)]platinum(II)
(3) dibromobis(trimethylphosphine)platinum(II)
(4) dibromodi(trimethylphosphine)platinum(II)

Pathfinders Academy JEE

CCC345

6. The total number of possible isomers for square-planar $[Pt(Cl)(NO_2)(NO_3)(SCN)]^{2\ominus}$ is :-
[JEE (Main) 2018]

(1) 16 (2) 8 (3) 24 (4) 12

CCC040

7. Which of the following complex has a possibility to exist as meridional isomer?
[JEE (Main) 2023]

(1) $[Co(NH_3)_3(NO_2)_3]$
(2) $[Co\ (en)_3]$
(3) $[Co\ (en)_2\ Cl_2]$
(4) $[Pt\ (NH_3)_2\ Cl_2]$

CCC317

8. Consider the following reaction and statements :
$[Co(NH_3)_4Br_2]^{\oplus}\ Br^{\ominus} \rightarrow [Co(NH_3)_3Br_3] + NH_3$
(I) Two isomers are produced if the reactant complex ion is a cis-isomer.
(II) Two isomers are produced if the reactant complex ion is a trans-isomer.
(III) Only one isomer is produced if the reactant complex ion is a trans-isomer.
(IV) Only one isomer is produced if the reactant complex ion is a cis-isomer.
The correct statements are :
[JEE (Main) 2018]

(1) (I) and (III) (2) (III) and (IV) (3) (II) and (IV) (4) (I) and (II)

CCC039

For More Join: @IITJEE_Advanced

9. The number of geometrical isomers possible in triamminetrinitrocobalt (III) is X and in trioxalatochromate (III) is Y. Then the value of X + Y is _____. **[JEE (Main) 2021]**

CCC047

10. Which one of the following complexes shows optical isomerism :- **[JEE (Main) 2016]**

(1) $[Co(NH_3)_4Cl_2]Cl$ (2) $[Co(NH_3)_3Cl_3]$

(3) $cis[Co(en)_2Cl_2]Cl$ (4) $trans[Co(en)_2Cl_2]Cl$

(en = ethylenediamine)

CCC033

11. The d-electronic configuration of an octahedral Co(II) complex having magnetic moment of 3.95 BM is : **[JEE (Main) 2025]**

(1) $t_{2g}^6 e_g^1$ (2) $t_{2g}^3 e_g^0$ (3) $t_{2g}^5 e_g^2$ (4) $e^4 t_2^3$

CCC384

12. Number of complexes from the following with even number of unpaired "d" electrons is___.

$[V(H_2O)_6]^{3\oplus}$, $[Cr(H_2O)_6]^{2\oplus}$, $[Fe(H_2O)_6]^{3\oplus}$, $[Ni(H_2O)_6]^{3\oplus}$, $[Cu(H_2O)_6]^{2\oplus}$

[Given atomic numbers : V = 23, Cr = 24, Fe = 26, Ni = 28, Cu = 29] **[JEE (Main) 2024]**

(1) 2 (2) 4 (3) 5 (4) 1

CCC346

13. The 'Spin only' Magnetic moment for $[Ni(NH_3)_6]^{2+}$ is_____$\times 10^{-1}$ BM.

(given = Atomic number of Ni : 28) **[JEE (Main) 2024]**

CCC347

14. The ratio of spin-only magnetic moment values $\mu_{eff}[Cr(CN)_6]^{3-} / \mu_{eff}[Cr(H_2O)_6]^{3+}$ is _____. **[JEE (Main) 2023]**

CCC318

15. $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ are respectively known as: **[JEE (Main) 2024]**

(1) Spin free Complex, Spin paired Complex

(2) Spin paired Complex, Spin free Complex

(3) Outer orbital Complex, Inner orbital Complex

(4) Inner orbital Complex, Spin paired Complex

CCC348

16. Match **List-I** with **List-II.**

List-I (Complex)		**List-II (Hybridisation & Magnetic characters)**	
(A)	$[MnBr_4]^{2-}$	(I)	d^2sp^3 & diamagnetic
(B)	$[FeF_6]^{3-}$	(II)	sp^3d^2 & paramagnetic
(C)	$[Co(C_2O_4)_3]^{3-}$	(III)	sp^3 & diamagnetic
(D)	$[Ni(CO)_4]$	(IV)	sp^3 & paramagnetic

Choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) (A)-(III), (B)-(II), (C)-(I), (D)-(IV) (2) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(3) (A)-(IV), (B)-(I), (C)-(II), (D)-(III) (4) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

CCC385

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For More Join: @IITJEE_Advanced

17. Identify the coordination complexes in which the central metal ion has d^4 configuration.

(A) $[FeO_4]^{2-}$ (B) $[Mn(CN)_6]^{3-}$ (C) $[Fe(CN)_6]^{3-}$ (D) $Cr_2(O-\overset{\overset{O}{\|}}{C}-Me)_4(H_2O)_2$

(E) $[NiF_6]^{2-}$

Choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) (C) and (E) only (2) (B), (C) and (D) only

(3) (B) and (D) only (4) (A), (B) and (E) only

CCC386

18. Total number of molecules/species from following which will be paramagnetic is _____.

O_2, O_2^+, O_2^-, NO, NO_2, CO, $K_2[NiCl_4]$,

$[Co(NH_3)_6]Cl_3$, $K_2[Ni(CN)_4]$ **[JEE (Main) 2025]**

CCC387

19. The correct order of spin-only magnetic moments among the following is :

(Atomic number : Mn = 25, Co = 27, Ni= 28, Zn = 30) **[JEE (Main) 2018]**

(1) $[ZnCl_4]^{2\ominus} > [NiCl_4]^{2\ominus} > [CoCl_4]^{2\ominus} > [MnCl_4]^{2\ominus}$

(2) $[CoCl_4]^{2\ominus} > [MnCl_4]^{2\ominus} > [NiCl_4]^{2\ominus} > [ZnCl_4]^{2\ominus}$

(3) $[MnCl_4]^{2\ominus} > [CoCl_4]^{2\ominus} > [NiCl_4]^{2\ominus} > [ZnCl_4]^{2\ominus}$

(4) $[NiCl_4]^{2\ominus} > [CoCl_4]^{2\ominus} > [MnCl_4]^{2\ominus} > [ZnCl_4]^{2\ominus}$

CCC041

20. The pair having the same magnetic moment is :-

[At. No.: Cr = 24, Mn = 25, Fe = 26, Co = 27] **[JEE (Main) 2016]**

(1) $[CoCl_4]^{2\ominus}$ and $[Fe(H_2O)_6]^{2\oplus}$ (2) $[Cr(H_2O)_6]^{2\oplus}$ and $[CoCl_4]^{2\ominus}$

(3) $[Cr(H_2O)_6]^{2\oplus}$ and $[Fe(H_2O)_6]^{2\oplus}$ (4) $[Mn(H_2O)_6]^{2\oplus}$ and $[Cr(H_2O)_6]^{2\oplus}$

CCC034

21. The correct increasing order of stability of the complexes based on Δ_o value is :

(I) $[Mn(CN)_6]^{3-}$ (II) $[Co(CN)_6]^{4-}$ (III) $[Fe(CN)_6]^{4-}$ (IV) $[Fe(CN)_6]^{3-}$

[JEE (Main) 2025]

(1) II < III < I < IV (2) IV < III < II < I (3) I < II < IV < III (4) III < II < IV < I

CCC388

22. The conditions and consequence that favours the t_{2g}^3, e_g^1 configuration in a metal complex are :

[JEE (Main) 2025]

(1) weak field ligand, high spin complex (2) strong field ligand, high spin complex

(3) strong field ligand, low spin complex (4) weak field ligand, low spin complex

CCC389

23. Identify the homoleptic complex(es) that is/are low spin.

(A) $[Fe(CN)_5NO]^{2-}$ (B) $[CoF_6]^{3-}$ (C) $[Fe(CN)_6]^{4-}$ (D) $[Co(NH_3)_6]^{3+}$

(E) $[Cr(H_2O)_6]^{2+}$

Choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) (B) and (E) only (2) (A) and (C) only (3) (C) and (D) only (4) (C) only

CCC390

24. In which of the following complexes the CFSE, Δ_0 will be equal to zero? **[JEE (Main) 2025]**

(1) $[Fe(NH_3)_6]Br_2$ (2) $[Fe(en)_3]Cl_3$ (3) $K_4[Fe(CN)_6]$ (4) $K_3[Fe(SCN)_6]$

CCC391

For More Join: @IITJEE_Advanced

25. From the magnetic behaviour of $[NiCl_4]^{2-}$ (paramagnetic) and $[Ni(CO)_4]$ (diamagnetic), choose the correct geometry and oxidation state. **[JEE (Main) 2025]**

(1) $[NiCl_4]^{2-}$: Ni^{II}, square planar, $[Ni(CO)_4]$: Ni(0), square planar
(2) $[NiCl_4]^{2-}$: Ni^{II}, tetrahedral, $[Ni(CO)_4]$: Ni(0), tetrahedral
(3) $[NiCl_4]^{2-}$: Ni^{II}, tetrahedral, $[Ni(CO)_4]$: Ni^{II}, square planar
(4) $[NiCl_4]^{2-}$: Ni(0), tetrahedral, $[Ni(CO)_4]$: Ni(0), square planar

CCC392

26. Match **List-I** with **List-II**.

List-I (Complex)		List-II (Hybridisation of central metal ion)	
(A)	$[CoF_6]^{3-}$	(I)	d^2sp^3
(B)	$[NiCl_4]^{2-}$	(II)	sp^3
(C)	$[Co(NH_3)_6]^{3+}$	(III)	sp^3d^2
(D)	$[Ni(CN)_4]^{2-}$	(IV)	dsp^2

Choose the ***correct*** answer from the options given below : **[JEE (Main) 2025]**

(1) (A)-(I), (B)-(IV), (C)-(III), (D)-(II) (2) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(3) (A)-(I), (B)-(II), (C)-(III), (D)-(IV) (4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

CCC393

27. The correct combination is **[JEE (Main) 2018]**

(1) $[Ni(CN)_4]^{2\ominus}$ – tetrahedral ; $[Ni(CO)_4]$ – paramagnetic
(2) $[NiCl_4]^{2\ominus}$– paramagnetic ; $[Ni(CO)_4]$ – tetrahedral
(3) $[NiCl_4]^{2\ominus}$ – diamagnetic ; $[Ni(CO)_4]$ – square – planar
(4) $[NiCl_4]^{2\ominus}$– square – planar ; $[Ni(CN)_4]^{2\ominus}$ – paramagnetic

CCC042

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28. The correct order of the following complexes in terms of their crystal field stabilization energies is : **[JEE (Main) 2025]**

(1) $[Co(NH_3)_4]^{2+}<[Co(NH_3)_6]^{2+}<[Co(en)_3]^{3+}<[Co(NH_3)_6]^{3+}$
(2) $[Co(NH_3)_4]^{2+}<[Co(NH_3)_6]^{2+}<[Co(NH_3)_6]^{3+}<[Co(en)_3]^{3+}$
(3) $[Co(NH_3)_6]^{2+}<[Co(NH_3)_6]^{3+}<[Co(NH_3)_4]^{2+}<[Co(en)_3]^{3+}$
(4) $[Co(en)_3]^{3+}<[Co(NH_3)_6]^{3+}<[Co(NH_3)_6]^{2+}<[Co(NH_3)_4]^{2+}$

CCC394

29. Consider the following complexes.

$[CoCl(NH_3)_5]^{2+}$, (A) $[Co(CN)_6]^{3-}$, (B)
$[Co(NH_3)_5(H_2O)]^{3+}$, (C) $[Cu(H_2O)_4]^{2+}$ (D)

The correct order of A, B, C and D in terms of wavenumber of light absorbed is : **[JEE (Main) 2024]**

(1) C < D < A < B (2) D < A < C < B
(3) A < C < B < D (4) B < C < A < D

CCC349

30. Arrange the following Cobalt complexes in the order of increasing Crystal Field Stabilization Energy (CFSE) value. **[JEE (Main) 2021]**

Complexes : $\underset{A}{[CoF_6]^{3\ominus}}$, $\underset{B}{[Co(H_2O)_6]^{2\oplus}}$, $\underset{C}{[Co(NH_3)_6]^{3\oplus}}$ and $\underset{D}{[Co(en)_3]^{3\oplus}}$

Choose the correct option :

(1) A < B < C < D (2) B < A < C < D (3) B < C < D < A (4) C < D < B < A

CCC048

For More Join: @IITJEE_Advanced

31. Which one of the following metal complexes is most stable? **[JEE (Main) 2021]**

(1) $[Co(en)(NH_3)_4]Cl_2$ (2) $[Co(en)_3]Cl_2$

(3) $[Co(en)_2(NH_3)_2]Cl_2$ (4) $[Co(NH_3)_6]Cl_2$

CCC046

32. Identify the **CORRECT** trend given below :

(Atomic No.= Ti : 22, Cr : 24 and Mo : 42) **[JEE (Main) 2016]**

(1) Δ_O of $[Cr(H_2O)_6]^{2\oplus} < [Mo(H_2O)_6]^{2\oplus}$ and Δ_O of $[Ti(H_2O)_6]^{3\oplus} < [Ti(H_2O)_6]^{2\oplus}$

(2) Δ_O of $[Cr(H_2O)_6]^{2\oplus} < [Mo(H_2O)_6]^{2\oplus}$ and Δ_O of $[Ti(H_2O)_6]^{3\oplus} > [Ti(H_2O)_6]^{2\oplus}$

(3) Δ_O of $[Cr(H_2O)_6]^{2\oplus} > [Mo(H_2O)_6]^{2\oplus}$ and Δ_O of $[Ti(H_2O)_6]^{3\oplus} > [Ti(H_2O)_6]^{2\oplus}$

(4) Δ_O of $[Cr(H_2O)_6]^{2\oplus} > [Mo(H_2O)_6]^{2\oplus}$ and Δ_O of $[Ti(H_2O)_6]^{3\oplus} < [Ti(H_2O)_6]^{2\oplus}$

CCC035

33. When Ethane-1,2-diamine is added progressively to an aqueous solution of Nickel (II) chloride, the sequence of colour change observed will be : **[JEE (Main) 2025]**

(1) Pale Blue → Blue → Green → Violet (2) Pale Blue → Blue → Violet → Green

(3) Green → Pale Blue → Blue → Violet (4) Violet → Blue → Pale Blue → Green

CCC395

34. Among the statements(a)-(d), the incorrect ones are-

(a) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments

(b) When $\Delta_0 < P$, the d-electron configuration of Co(III) in an octahedral complex is $t_{2g}^4 e_g^2$

(c) Wavelength of light absorbed by $[Co(en)_3]^{3\oplus}$ is lower than that of $[CoF_6]^{3\ominus}$

(d) If the Δ_0 for an octahedral complex of Co(III) is 18,000 cm^{-1}, the Δ_t for its tetrahedral complex with the same ligand will be 16,000 cm^{-1} **[JEE (Main) 2020]**

(1) (a) and (b) only (2) (c) and (d) only (3) (b) and (c) only (4) (a) and (d) only

CCC045

35. Two complexes, $[Cr(H_2O)_6]Cl_3$ (A) and $[Cr(NH_3)_6]Cl_3$ (B) are violet and yellow coloured, respectively. The incorrect statement regarding them is : **[JEE (Main) 2019]**

(1) Δ_0 value of (A) is less than that of (B).

(2) Δ_0 value of (A) and (B) are calculated from the energies of violet and yellow light, respectively

(3) Both absorb energies corresponding to their complementary colors.

(4) Both are paramagnetic with three unpaired electrons.

CCC044

36. $[Co_2(CO)_8]$ displays :- **[JEE (Main) 2017]**

(1) no Co-Co bond, four terminal CO and four bridging CO

(2) one Co-Co bond, six terminal CO and two bridging CO

(3) no Co–Co bond, six terminal CO and two bridging CO

(4) one Co-Co bond, four terminal CO and four bridging CO

CCC036

37. Low oxidation state of metals in their complexes are common when ligands: **[JEE (Main) 2022]**

(1) have good π -accepting character (2) have good σ -donor character

(3) are having good π -donating ability (4) are having poor σ -donating ability

CCC049

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. $EDTA^{4\ominus}$ is ethylenediaminetetraacetate ion. The total number of N–Co–O bond angles in $[Co(EDTA)]^{1\ominus}$ complex ion is **[JEE (Advanced) 2013]**

CCC073

2. Among $V(CO)_6$, $Cr(CO)_5$, $Cu(CO)_3$, $Mn(CO)_5$, $Fe(CO)_5$, $[Co(CO)_3]^{3-}$, $[Cr(CO)_4]^{4-}$, and $Ir(CO)_3$, the total number of species isoelectronic with $Ni(CO)_4$ is_____.
[Given, atomic number : V = 23, Cr = 24, Mn = 25, Fe = 26, Co = 27, Ni = 28, Cu = 29, Ir = 77]
[JEE (Advanced) 2024]

CCC350

3. The complex(es), which can exhibit the type of isomerism shown by $[Pt(NH_3)_2Br_2]$, is(are)
[en = $H_2NCH_2CH_2NH_2$] **[JEE (Advanced) 2023]**

(A) $[Pt(en)(SCN)_2]$ (B) $[Zn(NH_3)_2Cl_2]$
(C) $[Pt(NH_3)_2Cl_4]$ (D) $[Cr(en)_2(H_2O)(SO_4)]^+$

CCC320

4. The total number of possible isomers for $[Pt(NH_3)_4Cl_2]Br_2$ is _____ **[JEE (Advanced) 2021]**

CCC090

5. The pair(s) of coordination complex/ion exhibiting the same kind of isomerism is(are) – **[JEE (Advanced) 2013]**

(A) $[Cr(NH_3)_5Cl]Cl_2$ and $[Cr(NH_3)_4Cl_2]Cl$
(B) $[Co(NH_3)_4Cl_2]^{\oplus}$ and $[Pt(NH_3)_2(H_2O)Cl]^{\oplus}$
(C) $[CoBr_2Cl_2]^{2\ominus}$ and $[PtBr_2Cl_2]^{2\ominus}$
(D) $[Pt(NH_3)_3(NO_3)]$ Cl and $[Pt(NH_3)_3Cl]$ Br

CCC072

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6. Among the following options, select the option in which each complex in Set-I shows geometrical isomerism and the two complexes in Set-II are ionization isomers of each other.
[en = $H_2NCH_2CH_2NH_2$] **[JEE (Advanced) 2024]**

(A) Set-I : $[Ni(CO)_4]$ and $[PdCl_2(PPh_3)_2]$
Set-II : $[Co(NH_3)_5Cl]$ SO_4 and $[Co(NH_3)_5(SO_4)]Cl$
(B) Set-I : $[Co(en)(NH_3)_2Cl_2]$ and $[PdCl_2(PPh_3)_2]$
Set-II : $[Co(NH_3)_6]$ $[Cr(CN)_6]$ and $[Cr(NH_3)_6]$ $[Co(CN)_6]$
(C) Set-I : $[Co(NH_3)_3(NO_2)_3]$ and $[Co(en)_2Cl_2]$
Set-II : $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5(SO_4)]Cl$
(D) Set-I : $[Cr(NH_3)_5Cl]Cl_2$ and $[Co(en)(NH_3)_2Cl_2]$
Set-II : $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$

CCC351

7. Total number of cis N–Mn–Cl bond angles (that is, Mn–N and Mn–Cl bonds in cis positions) present in a molecule of cis-$[Mn(en)_2Cl_2]$ complex is ___ (en = $NH_2CH_2CH_2NH_2$)
[JEE (Advanced) 2019]

CCC086

8. The number of geometric isomers possible for the complex $[CoL_2Cl_2]^{\ominus}$ $(L = H_2NCH_2CH_2O^{\ominus})$ is
[JEE (Advanced) 2016]

CCC079

9. Among the complex ions,
$[Co(NH_2-CH_2-CH_2-NH_2)_2Cl_2]^{\oplus}$, $[CrCl_2(C_2O_4)_2]^{3\ominus}$, $[Fe(H_2O)_4(OH)_2]^{\oplus}$, $[Fe(NH_3)_2(CN)_4]^{\ominus}$, $[Co(NH_2\text{-}CH_2\text{-}CH_2\text{-}NH_2)_2(NH_3)Cl]^{2\oplus}$ and $[Co(NH_3)_4(H_2O)Cl]^{2\oplus}$, the number of complex ion(s) that show(s) cis-trans isomerism is - **[JEE (Advanced) 2015]**

CCC077

For More Join: @IITJEE_Advanced

10. Match each coordination compound in List-I with an appropriate pair of characteristics from List-II and select the correct answer using the code given below the lists. **[JEE (Advanced) 2014]**
{en = $H_2NCH_2CH_2NH_2$ ' atomic numbers ; Ti = 22 ; Cr = 24 ; Co = 27 ; Pt = 78}

List-I	List-II
(P) $[Cr(NH_3)_4Cl_2]Cl$	(1) Paramagnetic and exhibits ionisation isomerism
(Q) $[Ti(H_2O)_5Cl](NO_3)_2$	(2) Diamagnetic and exhibits cis-trans isomerism
(R) $[Pt(en)(NH_3)Cl]NO_3$	(3) Paramagnetic and exhibits cis-trans isomerism
(S) $[Co(NH_3)_4(NO_3)_2]NO_3$	(4) Diamagnetic and exhibits ionisation isomerism

Code :

	P	Q	R	S
(A)	4	2	3	1
(B)	3	1	4	2
(C)	2	1	3	4
(D)	1	3	4	2

CCC074

11. Among the following complexes, the total number of diamagnetic species is_____.
$[Mn(NH_3)_6]^{3+}$, $[MnCl_6]^{3-}$, $[FeF_6]^{3-}$, $[CoF_6]^{3-}$, $[Fe(NH_3)_6]^{3+}$, and $[Co(en)_3]^{3+}$
[Given, atomic number : Mn = 25, Fe = 26, Co = 27;
en = $H_2NCH_2CH_2NH_2$] **[JEE (Advanced) 2024]**
CCC352

12. LIST-I contains metal species and LIST-II contains their properties.

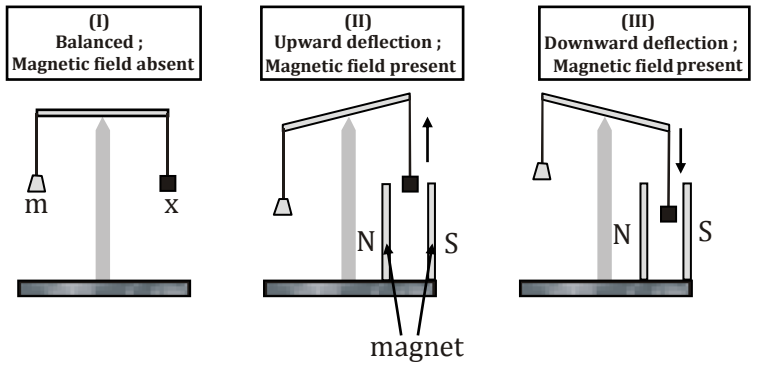
LIST-I	LIST-II
(I) $[Cr(CN)_6]^{4\ominus}$	(P) t_{2g} orbitals contain 4 electrons
(II) $[RuCl_6]^{2\ominus}$	(Q) μ(spin-only) = 4.9 BM
(III) $[Cr(H_2O)_6]^{2\oplus}$	(R) low spin complex ion
(IV) $[Fe(H_2O)_6]^{2\oplus}$	(S) metal ion in +4 oxidation state
	(T) d^4 species

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[Given : Atomic number of Cr = 24, Ru = 44, Fe = 26] **[JEE (Advanced) 2022]**
Metal each metal species in LIST-I with their properties in LIST-II, and choose the correct option
(A) I → R, T; II → P, S; III → Q, T; IV → P, Q (B) I → R, S; II → P, T; III → P, Q; IV → Q, T
(C) I → P, R; II → R, S; III → R, T; IV → P, T (D) I → Q, T; II → S, T; III → P, T; IV → Q, R
CCC092

13. The calculated spin only magnetic moments of $[Cr(NH_3)_6]^{3\oplus}$ and $[CuF_6]^{3\ominus}$ in BM, respectively, are
(Atomic numbers of Cr and Cu are 24 and 29, respectively) **[JEE (Advanced) 2021]**
(A) 3.87 and 2.84 (B) 4.90 and 1.73 (C) 3.87 and 1.73 (D) 4.90 and 2.84
CCC091

14. In an experiment, m grams of a compound X (gas/liquid/solid) taken in a container is loaded in a balance as shown in figure **I** below. In the presence of a magnetic field, the pan with **X** is either deflected upwards (figure **II**), or deflected downwards (figure **III**), depending on the compound **X**. Identify the correct statement (s) **[JEE (Advanced) 2020]**



For More Join: @IITJEE_Advanced

(A) If X is $H_2O(l)$, deflection of the pan is upwards.
(B) If X is $K_4[Fe(CN)_6](s)$, deflection of the pan is upwards.
(C) If X is $O_2(g)$, deflection of the pan is downwards.
(D) If X is $C_6H_6(l)$, deflection of the pan is downwards.

CCC088

15. Addition of excess aqueous ammonia to a pink coloured aqueous solution of $MCl_2 . 6H_2O$ (X) and NH_4Cl gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as 1 : 3 electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y.
Among the following options, which statements is(are) **CORRECT** ? **[JEE (Advanced) 2017]**
(A) The hybridization of the central metal ion in Y is d^2sp^3
(B) Z is tetrahedral complex
(C) Addition of silver nitrate to Y gives only two equivalents of silver chloride
(D) When X and Z are in equilibrium at 0°C, the colour of the solution is pink

CCC081

16. For the octahedral complexes of $Fe^{3\oplus}$ in $SCN^{\ominus}$ (thiocyanato – S)and in $CN^{\ominus}$ ligand environments, the difference between the spin only magnetic moments in Bohr magnetons (when approximated to the nearest integer) is : [Atomic number of Fe = 26] **[JEE (Advanced) 2015]**

CCC075

17. Consider the following complex ions P, Q and R,
P = $[FeF_6]^{3\ominus}$, **Q** = $[V(H_2O)_6]^{2\oplus}$ and **R** = $[Fe(H_2O)_6]^{2\oplus}$
The correct order of the complex ions, according to their spin-only magnetic moment values (in B.M.) is - **[JEE (Advanced) 2013]**
(A) R < Q < P (B) Q < R < P (C) R < P < Q (D) Q < P < R

CCC071

18. Match the electronic configurations in List-I with appropriate metal complex ions in List-II and choose the correct option.
[Atomic Number: Fe = 26, Mn = 25, Co = 27] **[JEE (Advanced) 2023]**

List-I		List-II	
(P)	$t_{2g}^6e_g^0$	(1)	$[Fe(H_2O)_6]^{2+}$
(Q)	$t_{2g}^3e_g^2$	(2)	$[Mn(H_2O)_6]^{2+}$
(R)	$e^2t_2^3s$	(3)	$[Co(NH_3)_6]^{3+}$
(S)	$t_{2g}^4e_g^2$	(4)	$[FeCl_4]^-$
		(5)	$[CoCl_4]^{2-}$

(A) P → 1; Q → 4; R → 2; S → 3
(B) P → 1; Q → 2; R → 4; S → 5
(C) P → 3; Q → 2; R → 5; S → 1
(D) P → 3; Q → 2; R → 4; S → 1

CCC319

19. The pair(s) of complexes wherein both exhibit tetrahedral geometry is(are)
(Note: py = pyridine **[JEE (Advanced) 2021]**
Given: Atomic numbers of Fe, Co, Ni and Cu are 26, 27, 28 and 29, respectively)
(A) $[FeCl_4]^-$ and $[Fe(CO)_4]^{2\ominus}$ (B) $[Co(CO)_4]^{\ominus}$ and $[CoCl_4]^{2\ominus}$
(C) $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2\ominus}$ (D) $[Cu(py)_4]^{\oplus}$ and $[Cu(CN)_4]^{3\ominus}$

CCC089

20. Choose the correct statement(s) among the following : **[JEE (Advanced) 2020]**
(A) $[FeCl_4]^{\ominus}$ has tetrahedral geometry.
(B) $[Co(en)(NH_3)_2Cl_2]^{\oplus}$ has 2 geometrical isomers.
(C) $[FeCl_4]^{\ominus}$ has higher spin – only magnetic moment than $[Co(en)(NH_3)_2Cl_2]^{\oplus}$.
(D) The cobalt ion in $[Co(en)(NH_3)_2Cl_2]$| has sp^3d^2 hybridization.

CCC087

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21. Among the species given below, the total number of diamagnetic species is___.
H atom, NO_2 monomer, $O_2^{\ominus}$ (superoxide), dimeric sulphur in vapour phase,
Mn_3O_4, $(NH_4)_2[FeCl_4]$, $(NH_4)_2[NiCl_4]$, K_2MnO_4, K_2CrO_4 **[JEE (Advanced) 2018]**
CCC085

22. Among $[I_3]^+$, $[SiO_4]^{4-}$, SO_2Cl_2, XeF_2, SF_4, ClF_3, $Ni(CO)_4$, XeO_2F_2, $[PtCl_4]^{2-}$, XeF_4, and $SOCl_2$, the total number of species having sp^3 hybridised central atom is _____. **[JEE (Advanced) 2023]**
CCC353

23. Match each set of hybrid orbitals from LIST-I with complex (es) given in LIST-II.
[JEE (Advanced) 2018]

LIST-I	**LIST-II**
(P) dsp^2	(1) $[FeF_6]^{4\ominus}$
(Q) sp^3	(2) $[Ti(H_2O)_3Cl_3]$
(R) sp^3d^2	(3) $[Cr(NH_3)_6]^{3\oplus}$
(S) d^2sp^3	(4) $[FeCl_4]^{2\ominus}$
	(5) $Ni(CO)_4$
	(6) $[Ni(CN)_4]^{2\ominus}$

The correct option is :-
(A) P → 5; Q → 4,6; R → 2,3; S → 1
(B) P → 5,6; Q → 4; R → 3; S → 1,2
(C) P → 6; Q → 4,5; R → 1; S → 2,3
(D) P → 4,6; Q → 5,6; R → 1,2; S → 3
CCC084

24. The geometries of the ammonia complexes of $Ni^{2\oplus}$, $Pt^{2\oplus}$ and $Zn^{2\oplus}$, respectively, are :
[JEE (Advanced) 2016]
(A) octahedral, square planar and tetrahederal
(B) square planar, octahederal and tetrahederal
(C) tetrahederal, square planar and octahederal
(D) octahederal, tetrahederal and square planar
CCC080

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25. Among $[Ni(CO)_4]$, $[NiCl_4]^{2\ominus}$, $[Co(NH_3)_4Cl_2]Cl$, $Na_3[CoF_6]$, Na_2O_2 and CsO_2, the total number of paramagnetic compounds is - **[JEE(Advanced) 2015]**
(A) 2 (B) 3 (C) 4 (D) 5
CCC078

26. The correct option(s) regarding the complex $[Co(en)(NH_3)_3(H_2O)]^{3\oplus}$:-
$(en = H_2NCH_2CH_2NH_2)$ is (are) **[JEE (Advanced) 2018]**
(A) It has two geometrical isomers
(B) It will have three geometrical isomers if bidentate 'en' is replaced by two cyanide ligands
(C) It is paramagnetic
(D) It absorbs light at longer wavelength as compared to $[Co(en)(NH_3)_4]^{3\oplus}$
CCC083

27. In the complex acetylbromidodicarbonylbis (triethyl phosphine) iron(II), the number of Fe–C bond(s) is- **[JEE (Advanced) 2015]**
CCC076

28. The correct statement(s) regarding the binary transition metal carbonyl compounds is (are)
(Atomic numbers : Fe = 26, Ni = 28) **[JEE (Advanced) 2018]**
(A) Total number of valence shell electrons at metal centre in $Fe(CO)_5$ or $Ni(CO)_4$ is 16
(B) These are predominantly low spin in nature
(C) Metal - carbon bond strengthens when the oxidation state of the metal is lowered
(D) The carbonyl C–O bond weakens when the oxidation state of the metal is increased
CCC082

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	C	B	C	B	C	D	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	D	A	C	B	A	B	D	D	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	A	D	D	B	A	C	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	A	C	B	A	B	B	D	B	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	D	D	A	B	C	A	B	D	D
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	A	A	B	A	D	B	D	B	B
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	D	B	B	C	D	B	A	D	D	B
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	C	D	C	A	C	C	A	C	C	B
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	B	C	D	D	D	B	C	C	B	D
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	C	D	A	B	A	C	A	C	D	A
Que.	101	102	103	104	105	106	107	108	109	110
Ans.	C	C	D	B	C	C	D	C	B	B
Que.	111	112	113	114	115	116	117	118	119	120
Ans.	A	D	A	D	A	B	B	A	C	D
Que.	121	122	123	124	125	126	127	128	129	130
Ans.	A	A	C	B	C	D	B	B	C	A
Que.	131	132	133	134	135	136	137	138	139	140
Ans.	B	A	D	B	C	C	D	B	A	C
Que.	141	142	143	144	145	146				
Ans.	C	B	D	D	C	B				

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABD	AB	AC	BC	ABCD	AC	CD	BCD	AB	AB
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	ABCD	AB	BD	AC	ACD	ABD	BD	ABC	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	B	C	D	B	C	A	C
Que.	31	32	33	34	35	36				
Ans.	D	C	A	A	B	A				

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EXERCISE (S)

Que.	1	2	3	4	5	6	7	8		
Ans.	3	34	5	4	2	33	36	(a) 4, (b) 9, (c) 4, (d) 4, (e) 5		
Que.	9	10	11	12	13	14	15	16	17	18
Ans.	3	0	2	15	2	9	2	2	6	5
Que.	19	20	21	22	23	24	25	26	27	28
Ans.	2	5	2	26.92	0.30	3.50	8.00	1.33	2.83	2.25
Que.	29	30								
Ans.	0.00	0.33								

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	14	1	1	1	3	4	1	1	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	28	1	2	4	3	6	3	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	1	3	4	2	2	2	2	2	2
Que.	31	32	33	34	35	36	37			
Ans.	2	2	3	4	2	2	1			

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	8	3	CD	6	BD	C	6	5	6	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	A	A	ABC	ABD	4	B	D	ABD	AC
Que.	21	22	23	24	25	26	27	28		
Ans.	1	5	C	A	B	ABD	3	BC		

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